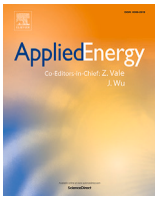
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A scalable mixed-integer programming model for multi-period hydropower planning in a single-reservoir system

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HIGHLIGHTS

- A tractable and realistic model for long-term hydropower operation planning.
- An approach that discretizes water levels, simplifies nonconvex dynamics, and prescribes intuitive operational curves for release guidance.
- Incorporates key features, including a head-dependent hydropower function, upstream flow travel time, smoothing constraints for water release, and a turbine switch-off policy.
- Validated using real-world data from the Lower Mekong Basin.

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ABSTRACT

This paper presents a mixed-integer linear formulation for solving a multi-period hydropower production planning problem in a single-reservoir system. The model offers a tractable yet realistic framework for long-term planning of hydropower operations. We utilize a production-driven dispatch model aimed at optimizing water releases at reservoirs and dams to maximize hydropower generation. The approach discretizes water levels, simplifies nonconvex dynamics, and prescribes easy-to-interpret operational curves for release guidance. Key features include a head-dependent hydropower function, upstream flow travel time, smoothing constraints on water release, and a turbine switch-off policy. The model is evaluated using real data from the Lower Mekong Basin.

1. Introduction

The rising reliance on fossil fuels over the past two decades has significantly intensified global air pollution [13]. Although fossil fuels are less expensive, they pose serious environmental risks [69]. To reconcile economic growth with environmental sustainability, the global energy community has increasingly promoted investment in renewables [69]. This transition supports better resource use and pollution reduction, aligning with the UN's "Sustainable Energy for All" initiative, which aims for universal access to sustainable energy by 2030. Developing countries are responding by adopting energy-efficient and cost-saving practices [60], including energy conservation practices in urban areas [60]. Renewable energy, particularly hydropower, has emerged as a key solution [13]. The global share of renewables in electricity generation rose from 28.1 % in 2021 to 30.2 % in 2023, driven largely by China [36]. Between 2023 and 2028, China is expected to deploy nearly four times more renewable capacity than the EU and five times more than the U.S., which will remain the second- and third-largest markets [36]. Hydropower, the world's largest renewable source, accounted for 14.2 % of global electricity in 2023, producing 4300 TWh, an increase of nearly 70 TWh (~2 %) over 2022 [36]. It is expected to retain its leading role among renewable electricity sources through 2028.

Hydropower generates electricity by releasing water from natural or human-made reservoirs through turbines. These reservoirs, typically located along rivers, store water to ensure a reliable supply. This storage allows hydropower plants to respond quickly to power demand, offering critical flexibility during peak periods [38]. Water may be stored during

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off-peak hours and released later as needed. Beyond energy generation, reservoirs support flood control, irrigation, navigation, recreation, and other ecosystem services [22]. Driven by these benefits and the increasing demand for renewable energy, reservoir construction has surged in developing countries. During the past decade, around 1000 dams have been built globally [49]. In the Mekong River Basin alone, there are 1055 dams—661 operational, 54 under construction, 331 planned, and a few canceled or decommissioned [7].

Hydropower production and ecosystem services are key factors in determining reservoir location and operation. Water storage and release decisions affect both energy output and the livelihoods of downstream communities [71]. Hydropower systems are typically classified as either storage-based or run-of-river (RoR). Storage-based systems use dams to control water levels for flexible power generation but can disrupt wildlife migration, flood habitats, and reduce agricultural and recreational river uses [53,72]. During the past two decades, RoR systems have been adopted as a lower-impact alternative to storage-based hydropower to reduce the environmental effects of dam construction and reservoir operations [9,38,41]. RoR systems feature minimal storage, with inflow closely matching the outflow, and aim to mimic natural river flow [51], allowing only small water level fluctuations. Water is typically diverted from higher elevations to a powerhouse (e.g., diversion or pondage schemes), or turbines are installed directly in the river (e.g., dam-toe schemes) [19]. RoR systems offer advantages over storage-based plants due to their minimal reservoir requirements, resulting in lower costs and shorter construction times [65]. They also mitigate common challenges such as inundation, sedimentation, fish migration barriers, navigation disruption, and community displacement [65], making them more socio-ecologically sustainable. However, their intermittent power output remains a key limitation [48], unlike storage systems that offer more stable generation. Few studies have addressed the optimization of RoR operations or developed models flexible enough to accommodate both regimes. For a review of existing RoR design models, see Tsuanyo et al. [64].

In hydropower plants, electricity generation depends on turbined release, net head, and turbine characteristics. This relationship, known as the hydropower production function (HPF), is typically modeled as a nonlinear function of water release and reservoir level. For tractability, it is often simplified to a univariate, head-independent function [71]. While this simplification may suffice in short-term planning, ignoring changes in net head, i.e. The height difference between reservoir and tailrace, can lead to inaccuracies [17]. Net head varies with sedimentation, upstream flows, reservoir aging, and other environmental factors, and neglecting it can result in suboptimal planning decisions [70]. When water is released through a turbine, some head is lost due to friction in the penstock, which depends on flow rate, pipeline length, and gravity [5]. While head-dependent HPFs have been proposed, their use in optimization is limited due to non-convexity. This paper presents a head-dependent HPF reformulation along with a scalable first-order linear approximation method, applicable to both storage and RoR hydropower operations.

We examine the optimal operation of a single-reservoir hydropower system over multiple periods. Like other renewables, hydropower involves environmental trade-offs. Even RoR systems can obstruct migratory routes for fish and some mammal species, while wind and solar energy require significant land use or increase bird and bat mortality. Given the inevitable role of hydropower in future renewable energy portfolios, especially in the developing world, it is essential to optimize energy production relative to environmental costs. This understanding is the critical first step toward developing a multi-objective framework to better evaluate and balance these tradeoffs.

The remainder of the paper is organized as follows. Section 2 reviews recent optimization models for hydropower systems and identifies existing challenges. Section 3 introduces the proposed hydropower production function, capturing the relationship between net head, water release, and power output, and presents a tractable reformulation for optimization. Section 4 describes the optimization model for scheduling operations in single-reservoir systems. In Section 5, we describe the construction of operational curves to address the tradeoff between hydrological uncertainty and hydropower production. Section 6 presents a case study using real data from the Sanakham project in the Lower Mekong Basin. Section 7 concludes with a summary of findings, model contributions, and directions for future research.

2. Literature review

Modeling hydropower operations is inherently complex due to the interplay of infrastructure limits, operational constraints, regulatory requirements (e.g., penstock capacity, reservoir levels, environmental flows) [8], and the unique characteristics of each project. This complexity is further amplified by limited data availability and the need to balance technical, economic, and environmental considerations. This paper advances existing approaches by integrating a head-dependent hydropower production function with key operational features into a scalable framework.

Realistic head-dependent HPFs depend on system-specific technical and operational characteristics. While the literature recognizes the importance of these features, simplifications are often necessary to maintain tractability. For example, Diniz and Maceira [24] assume constant head loss and use a piecewise-linear convex hull approximation of an HPF involving volume, release, spillage, and turbine characteristics. Similarly, Santos and Finardi [55] develop a large-scale Mixed-Integer Linear Programming (MILP) model using upper bounding hyperplanes and a Piecewise Convex Hull (PCH) formulation. Other approaches include modeling turbine performance under different head levels using piecewise linear curves [16], SOS2-based rectangular mesh schemes [71], and MILP formulations that incorporate grid-based approximations of efficiency functions [30]. While effective, these methods often require large number of binary variables, limiting scalability.

Li et al. [42] and Cheng et al. [17] address HPF nonlinearity using interpolation and discretized performance curves, potentially subject to large approximation errors. Tong et al. [63] use piecewise linear approximations to four one-dimensional nonlinear functions in an MILP model and propose a method to estimate and minimize the resulting approximation errors. In contrast, Daadaa et al. [21] propose a method to optimize over a set of pre-selected local efficiency points, maximizing hydropower production with a fixed upstream elevation based on the maximum reservoir volume. More recent work compares modeling strategies. Becker et al. [10] evaluate piecewise-linear vs. homotopy methods, with the former introducing discontinuities and the latter offering smoother but potentially suboptimal solutions. Guisandez and Perez-Diaz [31] address non-concave HPFs by discretizing head and discharge, comparing five MILP formulations, including univariate, rectangular, parallelogram, quadrilateral, and a logarithmic independent branching.

Some works embed the HPF modeling into larger models for broader hydropower systems with additional features. Daadaa et al. [21] present an MILP model for short-term hydropower unit commitment, aiming to maximize total production while optimizing water release, reservoir volume, and unit status under operational constraints. Tasseff et al. [61] address pump scheduling in water distribution networks, modeling the relationship between water flow, power production, and head loss, and handling HPF nonconvexity via polyhedral relaxations and piecewise linearization. Litlabo et al. [43] propose a short-term MILP model that prevents nonphysical overflow by linking spillage activation to reservoir volume using binary variables. Their model accounts for turbine efficiency, discharge, spillage, storage, and head loss. Each plant is represented as a single equivalent unit, aggregating multiple turbines to simplify notation and computation. Schäffer et al. [56] aggregates multiple hydropower plants into a single equivalent unit, with power production computed at the plant level using a piecewise-linear HPF based on segmented turbine discharge and efficiencies. Zhao et al. [71]

propose an MILP model for long-term planning, also aggregating plants into single units and using a two-variable production function based on average storage and total discharge.

Incorporating uncertainty in hydropower planning often focuses on inflow variability, using stochastic, probabilistic, and fuzzy approaches [4,20,29,37,57]. Fleten and Kristoffersen [27] develop a short-term stochastic program that improves operational efficiency using historical and forecast data. Favereau et al. [25] propose a multi-stage hydrothermal scheduling strategy using stochastic and robust optimization to address inflow, demand, and price uncertainties. Schäffer et al. [57] apply stochastic dynamic programming to evaluate trade-offs between hydropower flexibility and environmental constraints. Fuzzy logic is used by Ahmadianfar et al. [1] to model conflicting objectives under inflow uncertainty. Other studies focus on rule-based operations derived from historical data. Ahmadianfar et al. [3] introduce a self-adaptive algorithm (SATLDE) for generating reservoir release rules, while Ahmadianfar et al. [2] propose a linear rule linking release to storage, inflow, and upstream volume, with coefficients optimized within the model.

Various techniques are used to solve hydropower optimization problems, with MILP being the most common. Other methods include heuristics, dynamic programming, and decomposition techniques. For instance, Borghetti et al. [14] address unit commitment in a multi-unit pumped storage system using a mixed-integer nonlinear model with linearization to handle HPF non-convexity. Marcelino et al. [46] use a heuristic optimization method in Brazil to improve hydropower generation with lower water use. Evolutionary algorithms are widely used for handling nonlinear, nonconvex problems. Samadi-koucheksaraee et al. [54] compare Gradient Evolution with traditional optimization methods and show its effectiveness in complex systems with nonlinear HPFs, incorporating turbine efficiency and dynamic head levels depending on the reservoir's inflow, outflow, and evaporation. In multi-objective settings, Teegavarapu et al. [62] propose a fuzzy mixed-integer nonlinear model that balances hydropower production, water quality, and environmental releases. Vieira et al. [67] use a dual decomposition approach with a closed-form solution to maximize power generation while accounting for non-linear efficiency curves. Market-based models also appear in the literature. Mahan et al. [45] apply welfare maximization to develop a market price approach that leads to efficiency gains under different hydrological conditions. Pezalla and Obringer [50] develop a nonlinear model linking climate-driven electricity demand to generation strategies to prevent shortages during heatwaves.

Hydropower research increasingly incorporates economic objectives alongside technical and operational considerations. Avesani et al. [8] develop a short-term optimization model for a storage reservoir system to maximize revenue, using two econometric models to forecast hourly electricity prices based on past prices and forecast errors. Their analysis spans a full year, solving daily problems independently. Li et al. [42] propose an MILP model for unit commitment focused on minimizing operational costs, with decision variables for unit start-up/shutdown and water releases. The model accounts for net head variation and uses a three-dimensional interpolation to address HPF non-convexity. Similarly, Tong et al. [63] formulate a profit-maximizing MILP model incorporating hydropower revenues and start-up/shutdown costs. Steeger et al. [59] present a profit-maximization model for hydro producers with market power, optimizing bidding strategies under uncertain inflows, prices, and demand using Stochastic Dual Dynamic Programming (SDDP).

For a comprehensive review of hydropower scheduling models and algorithms, see Heintzmann et al. [34], which covers mathematical programming approaches such as MINLP, NLP, and heuristics including genetic algorithms (GA), simulated annealing (SA), and particle swarm optimization (PSO). The study highlights trade-offs between solution accuracy and computational cost in hydropower unit commitment. Additional surveys include Benkalai and Seguin [12], which outlines various hydropower optimization methods, and Tsuanyo1 et al. [65], which focuses on techno-economic models for RoR plant design. Ashofteh [11] review 76 studies (2002–2021) on dam reservoir optimization using exact, hybrid, and heuristic approaches.

Table 1 summarizes the key studies most relevant to this work, focusing on head-dependent HPFs and modeling of operational features. Columns 2–6 list technical aspects of the HPF, including turbine efficiency (Eff.), turbined release (Tr.), spillage (Spill.), storage volume (Vol.), and variable head loss (VHL) as a function of turbined release. These components are critical for capturing the performance and limitations of hydropower systems. A common limitation in the literature is model incompleteness, particularly regarding trade-offs between flood control and energy production. Flood control requires lowering

Table 1
Literature on the head-dependent HPF.

References	HPF features					Operational features			
	Eff.	Tr.	Spill.	Vol.	VHL	Time H.	T/R rules	Smooth.	Mov. T./Eco
[61]	✓	✓	✓	–	✓	Day (H)	–/–	–	–
[43]	✓	✓	✓	✓	✓	Day (H)	✓/–	–	–
[10]	✓	✓	–	✓	✓	Day (H)	✓/–	✓	–
[56]	✓	✓	–	–	–	Year (W)	✓/–	–	–
[55]	✓	✓	–	✓	✓	Day (H)	✓/–	–	–
[18]	✓	✓	–	–	–	Day (H)	✓/–	–	–
[21]	✓	✓	✓	✓	✓	Week (H)	✓/–	–	–
[71]	✓	✓	–	✓	✓	Year (M)	–/–	–	–
[52]	✓	✓	✓	✓	–	Month (D)	–/–	–	–
[16]	✓	✓	–	✓	–	Day (H)	✓/–	–	–
[40]	✓	✓	–	✓	–	Day (H)	–/–	–	–
[30]	✓	✓	✓	✓	✓	Day (H)	–/–	–	–
[23]	✓	✓	✓	✓	–	–	–/–	–	–
[17]	✓	✓	✓	✓	✓	Day (H)	–/–	–	–
[58]	✓	✓	–	–	–	Day (H)	–/–	–	–
[44]	✓	✓	–	✓	–	Week (D)	✓/–	–	–
[35]	✓	✓	–	–	–	Year (W)	–/–	–	–
[32]	✓	✓	–	–	–	Day (H)	–/–	–	–
[42]	✓	✓	✓	✓	✓	Day (H)	✓/–	–	–
[63]	✓	✓	✓	✓	✓	Day (H)	✓/–	–	–
[15]	✓	✓	–	✓	–	Week (H)	–/–	–	–
[14]	✓	✓	–	✓	–	Week (H)	–/–	–	–
[26]	✓	✓	✓	–	✓	Week (H)	✓/–	–	–
[28]	✓	✓	–	✓	–	Day (H)	–/–	–	–

reservoir levels ahead of upstream inflows, while hydropower benefits from higher water levels. Effective scheduling at daily or even hourly resolutions must balance these competing goals. Columns 7–10 in Table 1 capture operational features. Column "Time H" notes the planning horizon. This work falls into the long-term daily scheduling category (e.g., 365-day lookahead). Column "T/R" indicates whether turbine (T) and reservoir (R) operational rules are included ((✓) indicates presence and (–) absence). Turbine rules include start-up/shut-down policies, flow constraints, or level-based controls. Reservoir operation rules involve constraints to control water levels based on the relationship between inflow and outflow. Column "Smooth" reflects the presence of smoothing constraints to avoid unrealistic operational fluctuations. Column "mov.T/Eco" addresses anticipatory operations for upstream events and ecological considerations, often missing in traditional models. Traditionally, hydropower production is driven by dispatch demands, profit, and the availability of other power sources, while reservoir operations are based on hydro-meteorological conditions and operational rules [33,66]. However, sustainable planning must also account for ecosystem services. For example, Amjath-Babu et al. [6] propose a model that jointly optimizes hydropower, crop production, and flood risk reduction in water resource development.

This work addresses gaps identified in the literature by developing a tractable yet realistic mathematical programming model for hydropower operations. Our approach incorporates realistic HPF features and operational constraints into an MILP model designed for long-term daily planning. The model accommodates both traditional storage-based and RoR hydropower operations. We propose a hydropower function that depends on water release and net head, and we develop an approximation method to address the nonconvex nature of this function. Additionally, we introduce an operational curve that prescribes reservoir release decisions based on upstream conditions. This curve serves as both a user-friendly decision tool and a mechanism to integrate flood control and hydropower production decisions into a simple policy. The main contributions of our work are as follows:

- We propose a nonlinear HPF that integrates net head, turbined release, storage volume, and spillage. We then reformulate this function and introduce a first-order scheme to approximate the resulting HPF. Computational results show that this method provides a fast and high-quality approximation of the nonlinear HPF across a wide range of head levels, turbined releases, storage volumes, and spillage conditions.
- Our model incorporates features such as inflow travel time, accounting for the time required for water to reach the reservoir from upstream sources. We also implement smoothing constraints to prevent sudden or drastic changes in operational plans over time, such as fluctuations in water release or reservoir levels. Additionally, we include a turbine switch-off policy to be enforced under specific operational requirements.
- Our approach captures a continuous set of HPFs, each corresponding to a specific water head level. This approach allows for a finer search among operational plans compared to coarser methods that rely on a discrete set.
- We develop operational curves based on the optimal plans generated by the model for various upstream hydrology seasons. These curves account for hydrological uncertainty, flood control, and hydropower production, allowing the creation of easy-to-follow hydropower production plans.
- We embed the operational curve concept into an alternative MILP by imposing additional constraints. Our results indicate that the loss in hydropower generation is minimal compared to the reduction in uncertainty regarding future hydrology.
- Our model generates a multi-period release plan that maximizes hydropower production while accounting for inflow variability, storage dynamics, and operational constraints.

3. Hydropower production function (HPF)

The HPF proposed in this paper incorporates key features identified in the literature (Table 1), including turbine efficiency, reservoir water level as a function of storage volume, downstream level as a function of turbined release and spillage, and head loss dependent on turbined release. Our approach to constructing the HPF and addressing its nonconvexity improves the scalability of multi-period hydropower optimization models. We use the HPF in (1) from Diniz et al. [23], where hydropower production P is given by

$$P = \eta x \left[h_{up}(V) - h_{dn}(x+\Delta) - h_l(x)\right], \tag{1}$$

where η is a constant related to gravity, unit conversion, and efficiency of the power unit (generator and turbine), while V, x, and Δ represent the storage volume, turbined release, and spillage, respectively. The functions h_{up} and h_{dn} describe the upstream and downstream water levels given the storage volume and the total release (i.e., $x+\Delta$). The term in brackets is the net water head, which is the gross head (i.e., difference between h_{up} and h_{dn}) adjusted by the head loss h_l, which depends on the turbined release. Eqs. (2a) and (2b) specify polynomial functions for h_{up} and h_{dn}, where n, a_i, and b_i are estimated from sample data [23].

$$h_{up}(V) = \sum_{i=0}^{n} a_i V^i \tag{2a}$$

$$h_{dn}(x+\Delta) = \sum_{i=0}^{n} b_i (x+\Delta)^i \tag{2b}$$

$$h_l(x) = \frac{0.2083(100/c)^{1.852}}{d^{4.8655}} x^{1.852} \tag{2c}$$

We use the Hazen–Williams equation [5], as shown in (2c), to estimate head loss as a function of turbined release and other operational parameters. This equation captures the empirical relationship between water flow and the physical properties of the pipe to estimate frictional head loss. In Eq. (2c), $c > 0$ is the Hazen-Williams roughness constant, and d is the pipe's internal hydraulic diameter (in inches). The remaining constants are empirically determined.

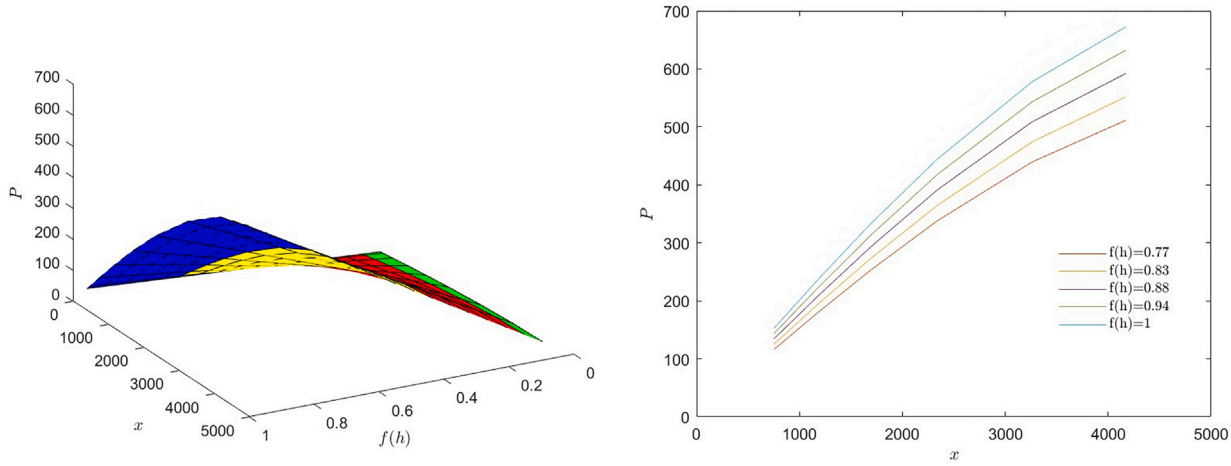
Eq. (1) typically models power production for a single unit. We apply it to a multi-unit plant by aggregating the units into a single equivalent unit. This reflects an *implicit* planning approach, where the number of active units is not explicitly modeled but can be inferred from total release and power output [23,24]. In this setting, the HPF represents the total power output of the plant. This approach is valid as long as Eq. (1) captures the plant's output for any combination of release, storage, and spillage.

In practice, Eq. (1) using (2a)–(2c) is a composition of a nonlinear concave function in V and x, and a nonlinear convex function on Δ. This structure is not desirable from a modeling perspective and results in a nonconvex function that is challenging to optimize [23]. To circumvent this challenge, we approximate Eq. (1) using

$$P = \eta x f(h)[h_{max} - C_L x^{\tau}], \tag{3}$$

where $C_L = \frac{0.2083(100/c)^{1.852}}{d^{4.8655}}$ and $\tau = 1.852$ [5]. The idea behind this approach is to calculate the hydropower production as a function of the maximum gross head h_{max} and then use a multiplicative correction factor $f(h)$ that depends on the head h to approximate the power loss when the water head is not at its maximum level. The correction factor $f(h)$ can also account for variable efficiency effects and adjust the HPF accordingly. This simplification reduces the number of arguments in the HPF (i.e., x and $f(h)$ only), improving its tractability when embedded into an optimization model.

Fig. 1a depicts the hydropower production in (3) as a function of the turbined release x and the correction factor $f(h)$. While the function is not concave, it consists of an infinite number of concave curves



(a) HPF as a function of turbined release and correction factor (b) HPFs for fixed values of $f(h)$

Fig. 1. Hydropower function (3).

on x for any given value of $f(h)$. Fig. 1b illustrates this observation by showing the HPF in (3) as a function of x for multiple values of $f(h)$. Proposition 3.1 formally proves this result examining the second derivative of function (3) with respect to x for a fixed value of $f(h)$.

Proposition 3.1. *Function (3) is concave on x for any given $f(h)$ and $x \geq 0$, whenever $0 \leq f(h) \leq 1$, $C_L \geq 0$, $|\tau| \geq 1$, and $\eta \geq 0$.*

Proof. The second derivative of P with respect to x is given by $\frac{\partial^2 P}{\partial x^2} = -(\tau+1)\tau\eta f(h) C_L x^{\tau-1}$. The conditions in the statement imply that $(\tau+1)\tau\eta f(h) C_L \geq 0$ and because $x \geq 0$, we obtain $\frac{\partial^2 P}{\partial x^2} \leq 0$. □

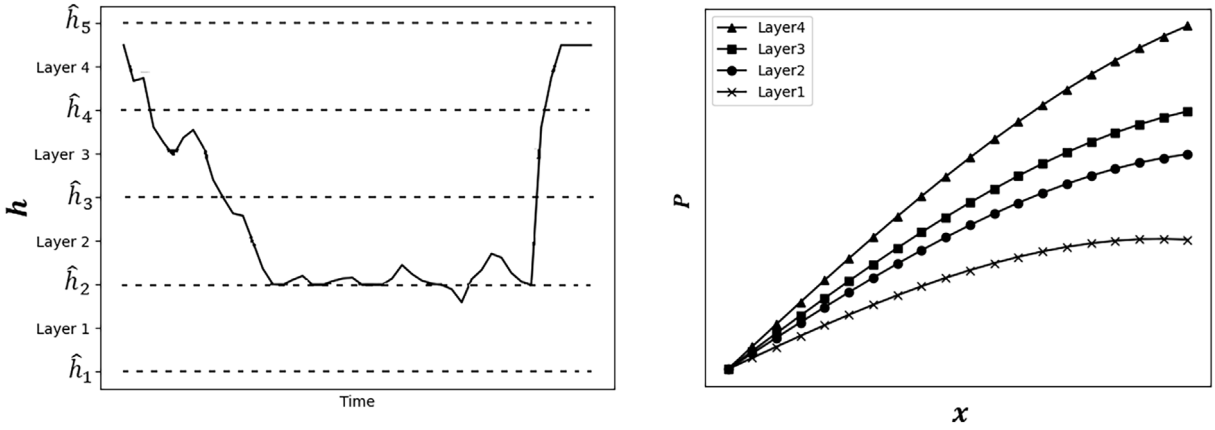
The conditions required by Proposition 3.1 are realistic as in typical hydropower systems we have $C_L \geq 0$, $\tau = 1.852$, $x \geq 0$, and $\eta \geq 0$.

To further simplify (3), we use sample values of $f(h)$, denoted by $f(\hat{h}_j)$ for the j-th sample. We approximate (3) by linearly interpolating between neighboring concave curves determined by the $f(\hat{h})$-values. The interpolation is performed using continuous variables λ_j. The resulting function remains nonconcave but is easier to describe using linear approximations. Mathematically, we first discretize the head domain as $\{\hat{h}_0, (\hat{h}_0, \hat{h}_1], (\hat{h}_1, \hat{h}_2], \ldots, (\hat{h}_{J-1}, \hat{h}_J]\}$, where $J > 0$, $\hat{h}_0 = 0$, $\hat{h}_J = h_{max}$, and $\hat{h}_0 < \hat{h}_1 < \ldots < \hat{h}_J$. We assume that for each value $\hat{h} \in \{\hat{h}_0, \hat{h}_1, \ldots, \hat{h}_J\}$, the value of $f(\hat{h})$ is known. Then, Eq. (3) can be reformulated as

$$P = \eta x [h_{max} - C_L x^{\tau}] \sum_{j=0}^{J} \lambda_j f(\hat{h}_j), \tag{4}$$

where $\lambda_j \in [0,1]$ for each j and $\sum_{j=0}^{J} \lambda_j = 1$ are used to interpolate between curves, and where only λ_j and λ_{j+1} are positive whenever $\hat{h}_j \leq h \leq \hat{h}_{j+1}$. Fig. 2 illustrates a four-layer water head discretization with their corresponding HPFs for a sample head value in each layer. The $\hat{h}$-values are the breakpoints of each layer. For each layer, only two λ-values are positive. According to Proposition 3.1, each combination of these λ-values induces a concave curve that follows (4). Although the figure illustrates a finite number of HPFs (one sample per layer), the number of curves that can be generated in each layer is infinite thanks to the continuous nature of the λ-values.

The quality of the HPF function in Eq. (3) compared to Eq. (1) depends on the correction factor $f(h)$. We estimate the parameters of this correction factor by minimizing the sum of squared errors between the estimated function in Eq. (3) and the exact function in Eq. (1). Details of this calculation and the performance of our approximation are elaborated in Section 6.1. Note that we have assumed a constant turbine efficiency (e.g., operation at the optimal or average efficiency level).



(a) Water head layers (b) HPF for sampled water head levels

Fig. 2. Example of a four-layer water head discretization.

However, the HPF in (1) can be extended to model η as a function of the release, $\eta(x)$. Consequently, the estimated correction factors $f(\hat{h}_j)$ in Equation (3) will also reflect adjustments for the variable efficiency for the given head level $\hat{h}_j$. Alternatively, a distinct efficiency value can be assigned to each net head interval, denoted by $\hat{\eta}_j$. In this case, the interpolation in Eq. (4) would use the terms $\lambda_j \hat{\eta}_j f(\hat{h}_j)$ in place of $\lambda_j f(\hat{h}_j)$ to account for variations in efficiency across net head intervals.

The proposed HPF in Eq. (4) presents the following advantages:

- By reformulating the multivariate HPF function in Eq. (1) as the interpolation of concave univariate functions, this approach provides a tractable method for optimizing hydropower production despite the nonconcave nature of the underlying function.
- Eq. (3) accommodates all continuous values of water release, rather than using a subset of values (e.g., local best efficiency points as in Daadaa et al. [21]). This improves the realism and accuracy of the optimization model. Moreover, our approach considers an infinite number of HPFs for each head interval, as opposed to a finite set (e.g., in Cheng et al. [17] and Li et al. [42]).
- Eq. (3) is a bi-variate function that directly calculates hydropower production based on turbined release and water head. As a result of Proposition 3.1, the proposed optimization model requires fewer binary variables to discretize all elements included in the HPF compared to existing piecewise linearization approaches (e.g., Tong et al. [63] and Guedes et al. [30]).
- The selection of λ-values is implicitly driven by the turbined release, spillage, and storage volume, and the interval where the net head h is located.

4. Single-reservoir optimization model

Optimization models are extensively utilized in reservoir management and hydropower production systems. These models typically include constraints related to water inventory balance, the hydropower production function, water level calculations, and bounds on water release and power production capacity. In our approach, we enhance these standard features by incorporating operational rules, smoothing constraints to prevent abrupt changes in release decisions, and accounting for travel time of upstream flow. Our model focuses on optimizing daily water release scheduling over a one-year planning horizon for a single reservoir. It prescribes optimal water releases and storage levels to maximize hydropower operations. Initially, the model is formulated as a mixed-integer nonlinear programming (MINLP) problem due to the nature of the hydropower production function (HPF) in (4). To reduce complexity and achieve near-optimal solutions, we linearize the model using first-order approximations, which simplify the problem while producing near-optimal operational plans.

To mathematically describe our problem, let T represent the set of time periods in the planning horizon. The reservoir can operate either as a traditional storage unit or as a RoR system with limited storage capacity, denoted by s_c. The water inventory balance within the reservoir is governed by the inflows and outflows. In the RoR operation system, the total inflow should be approximately equal to the total outflow during any given period. The total inflow at time t consists of inflow from upstream sources, I_t, and external inflow, such as precipitation, denoted by E_t. The total outflow is represented by the decision variable Q_t, which corresponds to the total release at time t. The total release Q_t consists of the flow directed to the turbine for hydropower production, x_t, and spillage, Δ_t. Spillage Δ_t occurs only when the release for hydropower x_t is at its maximum capacity or when the units are shut down. Additionally, to meet downstream water demands (e.g., for agriculture or navigability), a mandatory release m_t is required at time t.

We model the reservoir as the catchment area behind the dam (traditional storage) or turbines (RoR). The upstream water level is the water level at the dam or turbines, and the downstream water level is the water level after the power units. Upstream and downstream water levels at time t are defined by the decision variables $\hat{H}_t$ and $\check{H}_t$, respectively. The upstream water level is determined by the water volume, while the downstream water level depends on the total release. For operational purposes (e.g., turbine efficiency), our model enforces a maximum and minimum difference between the upstream and downstream water levels. This feature helps maintain operation within the equipment's design range, enhancing efficiency, environmental sustainability, and operational safety. We use (4) as the model's HPF and define $\beta_j = f(\hat{h}_j)$ as the breakpoint value for the correction factor between water head layers j and $j+1$. The details on the calculation of β_j are provided in Section 6.

The initial conditions of the system include the initial storage in the reservoir, denoted by s^0. This value is used in the water inventory balance constraint and the storage-water level curve to calculate the initial upstream water level. We define the parameter s^l as the minimum allowed storage and x^u as the maximum release to the turbine. The reservoir's storage at time $t \in T$ is denoted by s_t. The model includes a dynamic maximum total release that depends on the water level in the reservoir. The value of the maximum release, denoted by q_k, is determined by a set of threshold values of the reservoir's water level, denoted by $k = \{1, \ldots, |K|\}$. The appropriate threshold value is enforced as the water level changes, adjusting the maximum release.

Table 2 summarizes the sets, parameters, and decision variables used in this model including auxiliary parameters and decision variables needed in the constraints.

The proposed mixed-integer nonlinear model optimizes the objective function in (5), which is the total power produced by the reservoir throughout the planning horizon.

$$\max \quad \sum_{t \in T} P_t \tag{5}$$

The proposed modeling approach includes families of constraints to track the storage and water level, control the water release, and calculate the hydropower production. The mathematical formulation is generic and the model can be used for planning operations in storage-based or RoR systems.

Storage control:

Constraints (6) ensure a minimum volume of water in the reservoir during each time period while satisfying storage capacity limitations. The water storage is calculated in Constraints (7) using an inventory approach, where r is the upstream release period. This implies that it takes r periods of time for water to travel from upstream sources to the reservoir. We assume that the parameters $I_{1-r}, \ldots, I_{t-r}$ are known and represent a plausible forecast of the reservoir's inflow. Constraints (7) can be modified to reflect faster travel by reducing r in a subset of periods. Constraints (8) limit storage fluctuations to help protect aquatic habitats from abrupt water level changes, among other operational and environmental benefits. The parameter γ prevents the fast depletion of the reservoir and regulates the pace at which water is released. These constraints can be used to model a RoR operation by setting a small value for γ, as it determines the percentage of water that can be used in time period t with respect to storage in period $t-\tilde{t}$ $(t > \tilde{t})$. The parameter $\tilde{t}$ determines the past period used as a reference for release control.

$$s_t^l \le s_t \le s^c, \forall t \in T \tag{6}$$

$$s_t = \begin{cases} s^0 + I_{t-r} + E_t - Q_t & \text{t=1} \\ s_{t-1} + I_{t-r} + E_t - Q_t & \text{t>1} \end{cases} \tag{7}$$

$$s_t \ge (1-\gamma) s_{t-\tilde{t}}, \forall t \in T : t > \tilde{t} \tag{8}$$

Water release:

To implement the maximum total release constraints, the model uses a set of $|K|$ threshold values denoted as w_k, for $k \in K$, and a set of auxiliary binary variables y_{tk} determining whether the maximum release for

Table 2
Sets, parameters, and decision variables.

Sets:		Variables:	
T	Set of time periods	Q_t	Total water released at time t, $\forall t \in T$
J	Set of water head intervals	x_t	Water released for hydropower at time t, $\forall t \in T$
K	Set of water level threshold values to restrict release from the reservoir	Δ_t	Spillage released at time t, $\forall t \in T$
Parameters:			
I_t	Inflow from upstream sources at time t, $\forall t \in T$	s_t	Water volume stored at time t, $\forall t \in T$
E_t	External inflow at time t, $\forall t \in T$	P_t	Hydropower production at time t, $\forall t \in T$
u	Minimum difference allowed between upstream and downstream water levels	λ_{jt}	Coefficient for layer j at time t, $\forall t \in T, j \in J$
s_t^l	Minimum storage at time t, $\forall t \in T$	$\hat{H}_t$	Upstream water level at time t, $\forall t \in T$
s^0	Initial storage in the reservoir	$\check{H}_t$	Downstream water level at time t, $\forall t \in T$
s^c	Storage capacity for the reservoir		
m_t	Mandatory water release at time t (demand), $\forall t \in T$	d_t	Binary variable that equals 1 if the difference between upstream and downstream water volume at time t is less than u and equals 0 otherwise, $\forall t \in T$
$\hat{h}_j$	Breakpoint value for water head in layer j, $\forall j \in J$	θ_t	Binary variable that equals 1 if the water release to the turbine is at its maximum capacity and equals 0 otherwise, $\forall t \in T$
h	Gross head of the turbine	α_{jt}	Binary variable used to define λ_{jt} as a SOS2 variable, $\forall j \in J, t \in T$
c	Friction loss coefficient for the turbine	y_{tk}	Binary variable that equals 1 if the water level is less than w_k and equals 0 otherwise, $\forall t \in T, k \in K$
β_j	Breakout value for the correction factor between water head levels j and $j+1$, $\forall j \in J$		
g	Gravity constant		
a_t	Unit conversion parameter at time t, $\forall t \in T$		
e_t	Turbine efficiency at time t, $\forall t \in T$		
η_t	Constant related to gravity, unit conversion, and turbine efficiency at time t, $\forall t \in T$		
x^u	Maximum release to the turbine		
w_k	Threshold value for water level that restricts maximum release, $\forall k \in K$		
q_k	Maximum water release if the water level is less than w_k for water level k, $\forall k \in K$		
r	Number of periods for water to travel from upstream sources to the reservoir		
$\tilde{t}$	Number of periods during which the variations in the storage are controlled		
γ	The percentage of the water that can be used at time period t with respect to storage in period $t - \tilde{t}$		
ψ, ψ'	Allowed percentage of change in total and hydropower release between two consecutive time periods		

threshold k is enforced at time t. The maximum value of water release prevents channel erosion and other damages occurring in case of excessive release. Constraints (9) and (10) guarantee a release of no more than q_k when the upstream water level is less than or equal to the threshold value w_k. This helps mitigate channel erosion caused by excessive water releases. If the upstream water level is less than or equal to w_k in (10), then y_{tk} needs to be equal to zero, imposing a maximum value of water release q_k from the reservoir in (9). Note that the tolerance parameter ϵ helps enforcing $y_{kt} = 0$ when $\hat{H}_t = w_k$. The Big-M parameters can be defined as $M_k^1 = \max_{j \in K}\{q_j\} - q_k$, for each $k \in K$. We assume that $w_1 < w_2 < \ldots < w_{|K|}$, thus when the reservoir is at its maximum level at time t, i.e., $\hat{H}_t = w_{|K|}$, then $y_{t|k|} = 0$ from (10) and Q_t is restricted to be less than or equal to the maximum release $q_{|K|}$. If the reservoir is at its minimum level at time t, i.e., $\hat{H}_t = w_1$, then $Q_t \leq q_1$, with $q_1 = 0$ and no release is allowed. Constraints (11) impose a minimum release to satisfy downstream water demands for ecological needs, human consumption, and agriculture at time t. Constraints (12) indicate that the total release from the reservoir is the sum of the water release for hydropower and spillage. Constraints (13) and (14) prevent the total and hydropower releases from changing drastically between two consecutive time periods, where ψ and ψ' are parameters between 0 and 1 to capture the allowed percentage change. This enhances flow stability and helps alleviate ecological stress. Constraints (13) are inactive when the turbines are shut down and no water is released for hydropower. Constraints (15) set an upper bound on hydropower release based on plant capacity and ensure zero release when the turbines are shut down (i.e., when $d_t = 1$). Constraints (16) and (17) allow spillage when the release for hydropower is at its maximum capacity or when the turbines are shut down.

$$Q_t \leq q_k + M_k^1 y_{tk}, \forall t \in T, k \in K \tag{9}$$

$$\hat{H}_t \geq (w_k + \epsilon) y_{tk}, \forall t \in T, k \in K \tag{10}$$

$$Q_t \geq m_t, \forall t \in T \tag{11}$$

$$Q_t = \Delta_t + x_t, \forall t \in T \tag{12}$$

$$(1-\psi)x_{t-1} - x^u d_t \leq x_t \leq (1+\psi)x_{t-1} + x^u d_{t-1}, \forall t \in T \tag{13}$$

$$(1-\psi')Q_{t-1} \leq Q_t \leq (1+\psi')Q_{t-1}, \forall t \in T \tag{14}$$

$$x_t \leq x^u(1-d_t), \forall t \in T \tag{15}$$

$$x_t \geq x^u \theta_t, \forall t \in T \tag{16}$$

$$\Delta_t \leq q_{|K|}(\theta_t + d_t), \forall t \in T \tag{17}$$

Water level:

Constraints (18) state that the upstream water level is a function of the storage volume. In addition, Constraints (19) show that the downstream water level is calculated using a function of the release from the reservoir. The functions $\hat{f}$ and $\check{f}$ capture the physical properties of the

sites (e.g., topography) and are assumed to be estimated in advance. Section 6 discusses the estimation of these functions and their properties, and how they are incorporated into our model as they are unique to each river and reservoir.

$$\hat{H}_t = \hat{f}(s_t), \forall t \in T \tag{18}$$

$$\check{H}_t = \check{f}(Q_t), \forall t \in T \tag{19}$$

Hydropower production model:

The proposed modeling approach relies on a tractable approximation of the hydropower production dynamics using an HPF. The model's output is a hydropower production and release plan for the reservoir over the planning horizon. Constraints (20) capture the hydropower production function for each period, as presented in Section 3. In (20), we use $\eta_t = a_t e_t g$, where g is the gravity constant, and a_t and e_t are parameters for unit conversion and turbine efficiency at time t, respectively. Following the analysis from Section 3, the range of valid water head values is discretized into $|J|$ pre-specified intervals, each of which has its own HPF linking power output to the water release in the interval. We define $\hat{h}_j$ as the pre-calculated breakpoints in the water head's discretized domain.

For simplicity in the notation, we use $\beta_j = f(\hat{h}_j)$. Each β_j-coefficient in Constraints (20) is a correction factor corresponding to the water head breakpoint $\hat{h}_j$. To determine the value of β_j, we evaluate Eq. (1) using multiple values for water release levels and storage volumes along with their corresponding upstream and downstream water levels. Then, for gross heads falling within its range, we solve a least-squares problem to minimize the error between the results obtained from Eq. (1) and those obtained from Eq. (4). In this case, the β-parameters are the unknown values to estimate. Section 6 provides more details on this estimation.

Constraints (21) relate the calculated water head and a convex combination of two adjacent breakpoints in the water head domain, meaning that the water head value will be in an interval $[\hat{h}_{j-1}, \hat{h}_j]$ for some $j \in J : j > 1$. This requires that at most two of the λ-variables corresponding to adjacent layers should be positive. The binary variables α_{jt} associated with each layer determine these two variables through Constraints (22)–(24). The binary variable α_{jt} is equal to 1 if the water head lies between $\hat{h}_{j-1}$ and $\hat{h}_j$ for $j \in \{1, \dots, |J|\}$. For example, if $\alpha_{0t} = 1$, then the water head at time t is found as the convex combination of $\hat{h}_0$ and $\hat{h}_1$, using weights λ_{0t} and λ_{1t} (i.e., $\hat{H}_t - \check{H}_t = \lambda_{0t}\hat{h}_0 + \lambda_{1t}\hat{h}_1$). This is achieved by imposing that exactly one of the α-variables must be equal to 1 (Constraints (25)), and that the sum of the λ-variables must be equal to 1 (Constraints (26)) for every period. Constraints (27) and (28) track the difference between water levels at the reservoir (upstream) and downstream and help enforce the powerhouse shut-down policy when this difference is less than or equal to u. Constraints (29) switch off the turbine when the difference in the water levels is less than the pre-specified value u. The big-M parameter in Constraints (27) can be defined as $M^2 = h_{max} - u$. The big-M parameter M^3 in Constraints (29) is the hydropower capacity of the turbine. Constraints (30)–(34) define the nature of the decision variables.

$$P_t \le \eta_t x_t \left(h_{max} - C_L x_t^{\tau}\right) \sum_{j \in J} \lambda_{jt}\beta_j, \forall t \in T \tag{20}$$

$$\hat{H}_t - \check{H}_t = \sum_{j \in J} \hat{h}_j \lambda_{jt}, \forall t \in T \tag{21}$$

$$\lambda_{0t} \le \alpha_{0t}, \forall t \in T \tag{22}$$

$$\lambda_{jt} \le \alpha_{j-1,t} + \alpha_{jt}, \forall j \in \{1, \dots, |J|-1\}, t \in T \tag{23}$$

$$\lambda_{|J|,t} \le \alpha_{|J|-1,t}, \forall t \in T \tag{24}$$

$$\sum_{j \in J} \alpha_{jt} = 1, \forall t \in T \tag{25}$$

$$\sum_{j \in J} \lambda_{jt} = 1, \forall t \in T \tag{26}$$

$$\hat{H}_t - \check{H}_t \le u + M^2(1 - d_t), \forall t \in T \tag{27}$$

$$\hat{H}_t - \check{H}_t \ge u\left(1 - d_t\right), \forall t \in T \tag{28}$$

$$P_t \le M^3(1 - d_t), \forall t \in T \tag{29}$$

$$Q_t, x_t, \Delta_t, s_t, P_t \ge 0, \forall t \in T \tag{30}$$

$$\lambda_{jt} \in [0,1], \forall j \in J, t \in T \tag{31}$$

$$d_t, \theta_t \in \{0,1\}, \forall t \in T \tag{32}$$

$$\alpha_{jt} \in \{0,1\}, \forall j \in J, t \in T \tag{33}$$

$$y_{tk} \in \{0,1\}, \forall t \in T, k \in K \tag{34}$$

We propose a first-order approximation to circumvent the nonconvexity caused by the hydropower production calculation in Constraints (20). Proposition 4.1 formalizes the proposed first-order approximation approach. Although the hydropower function in (20) is neither concave nor convex (see Fig. 1a), the motivation for using a first-order approximation is supported by a drastic reduction in problem complexity and the resulting mathematical properties. The concave function $\tilde{f}(\mathbf{x})$ from Proposition 4.1 is tractable for a maximization problem and can be used to approximate P_t from (20). Moreover, at most two λ-variables can be positive at any time, and they belong to the interval $[0, 1]$, which limits the first-order approximation error. This is because the hyperplanes on the right-hand side of (35) will have a limited range of applicability, determined by the λ-variables.

Proposition 4.1. *Let $f(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ be a concave function, and let $\{\mathbf{x}_1, \mathbf{x}_2, .., \mathbf{x}_l\}$ be a set of points in the function domain. The function $\tilde{f}(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ given by*

$$\tilde{f}(\mathbf{x}) = \min_{i=1,\dots,l} \{f(\mathbf{x}_i) + \nabla f(\mathbf{x}_i)^T(\mathbf{x} - \mathbf{x}_i)\}, \tag{35}$$

where $\nabla f(\mathbf{x})$ is the gradient of $f(\mathbf{x})$ evaluated at point $\mathbf{x}$, satisfies $f(\mathbf{x}) \le \tilde{f}(\mathbf{x})$ for any $\mathbf{x} \in \mathbb{R}^n$.

To reformulate Constraints (20), we define Ω as the set of sampling points $(\overline{x}^{\omega}, \overline{\lambda}^{\omega})$ for each $\omega \in \Omega$. For each t and sampling point ω, the hydropower production dynamics are reformulated in Constraints (36).

$$\begin{aligned} P_t \le{}& \eta_t \overline{x}_t^{\omega}\left([h_{max} - C_L(\overline{x}_t^{\omega})^{\tau}]\right) \sum_{j \in J} \overline{\lambda}_{jt}^{\omega}\beta_j \\ &+ \begin{pmatrix} \eta_t \sum_{j \in J} \overline{\lambda}_{jt}^{\omega}\beta_j \left[h_{max} - C_L(\tau+1)(\overline{x}_t^{\omega})^{\tau}\right] \\ \beta_1 \eta_t \overline{x}_t^{\omega}\left[h_{max} - C_L(\overline{x}_t^{\omega})^{\tau}\right] \\ \dots \\ \beta_{|J|}\eta_t \overline{x}_t^{\omega}\left[h_{max} - C_L(\overline{x}_t^{\omega})^{\tau}\right] \end{pmatrix}^T \begin{pmatrix} x_t - \overline{x}_t^{\omega} \\ \lambda_{1t} - \overline{\lambda}_{1\,t}^{\omega} \\ \dots \\ \lambda_{|J|t} - \overline{\lambda}_{|J|t}^{\omega} \end{pmatrix}, \\ &\forall t \in T, \omega \in \Omega \end{aligned} \tag{36}$$

This linearization approach does not require any additional binary variables, and the approximation quality can be controlled by adjusting the number of sampling points and the number of water head layers. Empirical tests show that the first-order approximation is effective and provides hydropower values that are close to the exact HPF at optimal and non-optimal water release values. Section 6 provides further details on these results.

Additional model features:

The proposed formulation is based on a *production-driven* dispatch model, where the objective is to optimize water releases to maximize hydropower generation. However, the model can be adapted to alternative dispatch strategies. For instance, it can be converted into a *price-driven*

dispatch model by incorporating time-dependent prices into the objective function–i.e., the objective function in (5) can be modified to $\sum_{t\in T} \pi_t P_t$, where π_t is the selling price per MW in period t, assuming a price forecast is available. Similarly, the maximization model can be adapted to a *demand-driven* dispatch framework by introducing constraints that limit production to what the market can absorb–i.e., adding $P_t \le D_t$ to the model, where D_t is the power demand in period t. In cases where $P_t < D_t$, unmet demand would need to be covered by alternative sources (e.g., thermal generation).

Constraints (27)–(29) enforce the turbine switch-off policy when the water head falls below a threshold u. This approach can be extended to include additional interval-based *forbidden zones*, where the water release must remain within specific bounds for given head levels. Formally, let $\Gamma = \{[\check{\ell}_1, \hat{\ell}_1], \dots, [\check{\ell}_{|\Gamma|}, \hat{\ell}_{|\Gamma|}]\}$ denote the set of water head intervals that trigger release constraints, where $\check{\ell}_i < \hat{\ell}_i$, for each $i \in \Gamma$ and $\hat{\ell}_i < \check{\ell}_{i+1}$, for $i = 1, \dots, |\Gamma| - 1$. Let Γ^c be the set of intervals such that $\Gamma \cup \Gamma^c = [0, h_{max}]$, i.e., if $[\check{\ell}_i, \hat{\ell}_i]$ and $[\check{\ell}_{i+1}, \hat{\ell}_{i+1}]$ belong to Γ, then $[\hat{\ell}_i, \check{\ell}_{i+1}] \in \Gamma^c$. Set Γ^c contains the intervals $[\hat{\ell}_0 = 0, \check{\ell}_1]$ if $\check{\ell}_1 > 0$ and $[\hat{\ell}_{|\Gamma|}, \check{\ell}_{|\Gamma|+1} = h_{max}]$ if $\hat{\ell}_{|\Gamma|} < h_{max}$. For each $[\check{\ell}_i, \hat{\ell}_i] \in \Gamma$, define $\Lambda_i = \{[\check{x}_{1i}, \hat{x}_{1i}], \dots, [\check{x}_{|\Lambda_i|,i}, \hat{x}_{|\Lambda_i|,i}]\}$ as the set of admissible release intervals associated with head interval $i \in \Gamma$. This implies that if the net head $\hat{H}_t - \check{H}_t \in [\check{\ell}_i, \hat{\ell}_i]$, then the release must satisfy $x_t \in [\check{x}_{1i}, \hat{x}_{1i}] \vee \dots \vee [\check{x}_{|\Lambda_i|,i}, \hat{x}_{|\Lambda_i|,i}]$. To implement these conditions using linear constraints, we define binary variables z_{it}, which equal one if the head at time t belongs to the interval $[\check{\ell}_i, \hat{\ell}_i]$, and binary variables $\bar{d}_{pit}$, which equal one if the release at time t lies in interval $[\check{x}_{pi}, \hat{x}_{pi}]$, for all $i \in \Gamma$ and $p \in \Lambda_i$. Constraints (37), (38), and (40) identify the interval to which the water head at time t belongs. If it belongs to none of the intervals in Γ, then $z_{it} = 0$ for each $i \in \Gamma$. Constraints (39) and (41) enforce that if the water head belongs to interval $i \in \Gamma$ at time t (i.e., $z_{it} = 1$), then the release at time t must belong to an admissible interval (i.e., $\exists p \in \Lambda_i : \bar{d}_{pit} = 1$). No release constraint is enforced if the water head belongs to an interval in Γ^c. The forbidden zones feature potentially introduces a large number of additional binary variables and constraints, depending on the number of intervals. However, these constraints can be incorporated efficiently using a Branch-and-Cut approach, where they are checked at each integer solution during the solution process and added only when violated.

$$\check{\ell}_i - x^u(1 - z_{it}) \le \hat{H}_t - \check{H}_t \le \hat{\ell}_i + x^u(1 - z_{it}) \,, \forall i \in \Gamma, t \in T \tag{37}$$

$$\hat{\ell}_i - x^u(1 - z_{it}) \le \hat{H}_t - \check{H}_t \le \check{\ell}_i + x^u(1 - z_{it}) \,, \forall i \in \Gamma^c, t \in T \tag{38}$$

$$\check{x}_{pi} - x^u(1 - \bar{d}_{pit}) \le x_t \le \hat{x}_{pi} + x^u(1 - \bar{d}_{pit}) \,, \forall i \in \Gamma, p \in \Lambda_i, t \in T \tag{39}$$

$$\sum_{i \in \Gamma \cup \Gamma^c} z_{it} = 1, \ \forall t \in T \tag{40}$$

$$\sum_{p \in \Lambda_i} \bar{d}_{pit} \ge z_{it}, \ \forall i \in \Gamma, t \in T \tag{41}$$

Appendix A presents the model size—excluding additional features—in terms of the number of variables and constraints.

5. Operational curve

Reservoir operators adjust gates to meet target releases and maintain water levels within physical and technical guidelines, including flood control, conservation, and buffer zones [39,68]. The flood zone accommodates runoff, the conservation zone maintains optimal supply levels, and the buffer zone secures water during dry or high-demand periods. Operational curves, pre-calculated from observed data, guide decisions without requiring complex models. Fig. 3 shows a piecewise linear operational curve linking inflow and reservoir level, where each segment reflects a specific operational regime with a buffer range between minimum and maximum levels. The operational curve reflects different trends across inflow values due to flow travel time. After heavy upstream rainfall, water takes time to reach the reservoir, depending on distance, channel properties, and flow velocity. In Fig. 3, the x-axis accounts for this delay. Anticipating inflows based on travel time enables operators to pre-release water, creating buffer storage and preventing overflow. In Zone 1 of Fig. 3, upstream flow is within normal limits, keeping the reservoir level stable with no spillage. In Zone 2, rising inflows trigger controlled releases to manage anticipated flooding and flush sediment, lowering the reservoir level. Zone 3 captures the peak flood phase, where inflow exceeds outflow despite maximum release, causing the reservoir level to rise until it returns to normal.

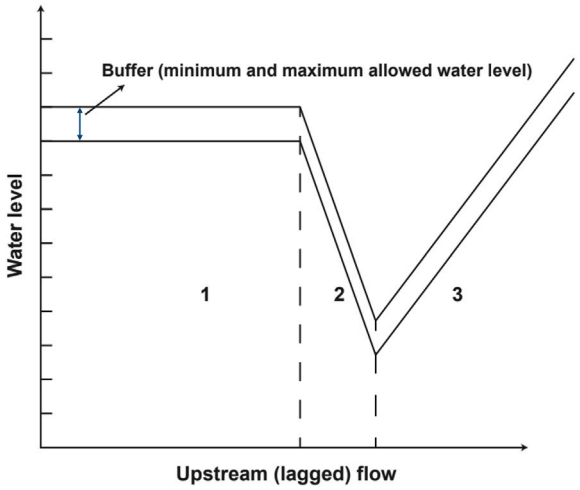


Fig. 3. Operational curve concept.

The operational curve serves as a practical tool for reservoir management by reducing risk and addressing hydrological uncertainty. It enables operators to proactively manage water levels based on upstream flows, offering a buffer against unexpected inflows and minimizing flood risks. By providing clear guidance, the curve supports informed decision-making, helping to stabilize reservoir operations and avoid abrupt changes in water levels and hydropower output. Unlike predictive optimization models, the operational curve offers a conservative, rule-based approach that is easy-to-use even under uncertain hydrological conditions.

We construct a *data-driven* operational curve by learning from previous operational schedules. To derive the curve, the hydropower model in (5)–(34) is solved using N years of daily input data covering various hydrological conditions. The water release-water level data are then input into a *fitting* optimization model designed to identify the breakpoints and line segments that define the operational curve (as in Fig. 3). The model's goal is to generate an operational curve that maximizes the number of optimal plans within the water levels specified by the curve. In Section 6, we compare the data-driven operational curve with a *technical* curve developed by experts, which is based on hydrological simulations rather than derived from a sample of optimal operational plans. Furthermore, in Section 6, we quantify the differences in hydropower production between the results from the model in (5)–(34) and those based on the operational curve.

Mathematically, let C be the set of observed pairs (lagged upstream flow, water level), with coordinates (x_c, y_c) for each $c \in C$. Let $\mathcal{P}$ denote the set of characteristic segments of the curve. The model determines the breakpoints for the x-axis, for $p \in \mathcal{P}$. Additionally, the model calculates the slope and intercept of each line segment connecting these breakpoints, represented by a_p and b_p, respectively. We incorporate a buffer, ρ, for all segments to allow the water level to vary within a range

Sets:
C: set of all points in the input water release-water level plot
$\mathcal{P}$: set of reservoir characteristic segments

Parameters:
(x_c, y_c): coordinates of the input points in the water release-water level plot, sorted in non-decreasing order of x_c,$\forall c \in C$
ρ: buffer value
d_p^l: minimum value for the x-coordinate of the breaking point between segment p and $p+1$, $\forall p \in \mathcal{P} : p < |\mathcal{P}|$
d_p^u: maximum value for the x-coordinate of the breaking point between segment p and $p+1$, $\forall p \in \mathcal{P} : p < |\mathcal{P}|$

Decision variables:
δ_{cp}: binary variable that equals 1 if point c is in segment p; 0 otherwise, $\forall c \in C, p \in \mathcal{P}$
σ_{cp}: binary variable that equals 1 if the x-coordinate of point c is the breakpoint between segments p and $p+1$; 0 otherwise, $\forall c \in C, p \in \mathcal{P} : p < |\mathcal{P}|$
b_p: slope of line segment p, $\forall p \in \mathcal{P}$
a_p: intercept of line segment p, $\forall p \in \mathcal{P}$

of values for any given release amount, rather than requiring a specific value. To maximize the number of optimal points that adhere to the curve, we introduce a binary variable δ_{cp}, where $\delta_{cp} = 1$ if point c lies within segment p, and $\delta_{cp} = 0$ otherwise. Moreover, we introduce the binary variable σ_{cp}, which equals 1 if the x-coordinate of point $c \in C$ is selected as the breakpoint between segments p and $p+1$, and 0 otherwise. Continuity of the operational curve is required at each breakpoint. The resulting optimization model produces an operational curve with $|\mathcal{P}|$ segments.

Optimization model:

$$\max \sum_{p\in\mathcal{P}} \sum_{c\in C} \delta_{cp} \tag{42}$$

$$\text{s.t.} \quad a_p + b_p x_c - M^4(1-\delta_{cp}) \le y_c \le a_p + b_p x_c + \rho + M^4(1-\delta_{cp}), \quad \forall c \in C, p \in \mathcal{P} \tag{43}$$

$$\sum_{c\in C} x_c \sigma_{cp} - M^5(1-\delta_{cp}) \le x_c \le \sum_{c\in C} x_c \sigma_{c,p+1} + M^5(1-\delta_{cp}), \quad \forall c \in C, p \in \mathcal{P}, p < |\mathcal{P}| \tag{44}$$

$$a_{p+1} + b_{p+1} x_c - M^4(1-\sigma_{cp}) \le a_p + b_p x_c \le a_{p+1} + b_{p+1} x_c + M^4(1-\sigma_{cp}), \forall c \in C, p \in \mathcal{P}, p < |\mathcal{P}| \tag{45}$$

$$\sum_{p\in\mathcal{P}} \delta_{cp} \le 1, \forall c \in C \tag{46}$$

$$\sum_{c\in C} \sigma_{cp} = 1, \forall p \in \mathcal{P} : p < |\mathcal{P}| \tag{47}$$

$$\sigma_{cp} \le \delta_{cp}, \forall c \in C, p \in \mathcal{P} \tag{48}$$

$$d_p^l \le \sum_{c\in C} c_c \sigma_{cp} \le d_p^u, \forall p \in \mathcal{P} : p < |\mathcal{P}| \tag{49}$$

$$\delta_{cp} \in \{0,1\}, \forall p \in \mathcal{P}, c \in C \tag{50}$$

$$\sigma_{cp} \in \{0,1\}, \forall p \in \mathcal{P}, c \in C \tag{51}$$

The objective in (42) maximizes the number of points inside the operational curve with buffer ρ and using line segments defined by a_p, b_p, and σ_{cp}. Constraints (43) ensure that if point c lies on segment p, its y-value stays within the band $[b_p x_c + a_p, b_p x_c + a_p + \rho]$. Here, M^4 is a Big-M constant equal to the reservoir's maximum water level. Constraints (44) preserve segment consistency by ensuring that breakpoints lie between the x-values of adjacent characteristic segments, using M^5 as the maximum release value. Constraints (45) ensure the continuity of the operational curve at its breakpoints, and Constraints (46) enforce that no point can be assigned to more than one segment of the operational curve. Constraints (47) enforce that there is only one point selected for each breakpoint. Constraints (48) enforce that if a point is selected as a breakpoint between segments p and $p+1$, then it must belong to segment p. Constraints (49) imposes limits on the x-coordinate of the breakpoint within each segment, specifically aligned with the shape of upstream inflow-water level scatter plot. Constraints (50) and (51) describe the binary nature of the δ- and σ-variables.

6. Case study and numerical results

The Mekong River spans approximately 4880 km [47] and flows through six countries–China and Myanmar in the Upper Mekong Basin, and Laos, Thailand, Cambodia, and Vietnam in the Lower Mekong Basin. Due to its favorable hydrological conditions, the river is heavily dammed, with 1055 dams: 661 operational, 54 under construction, 331

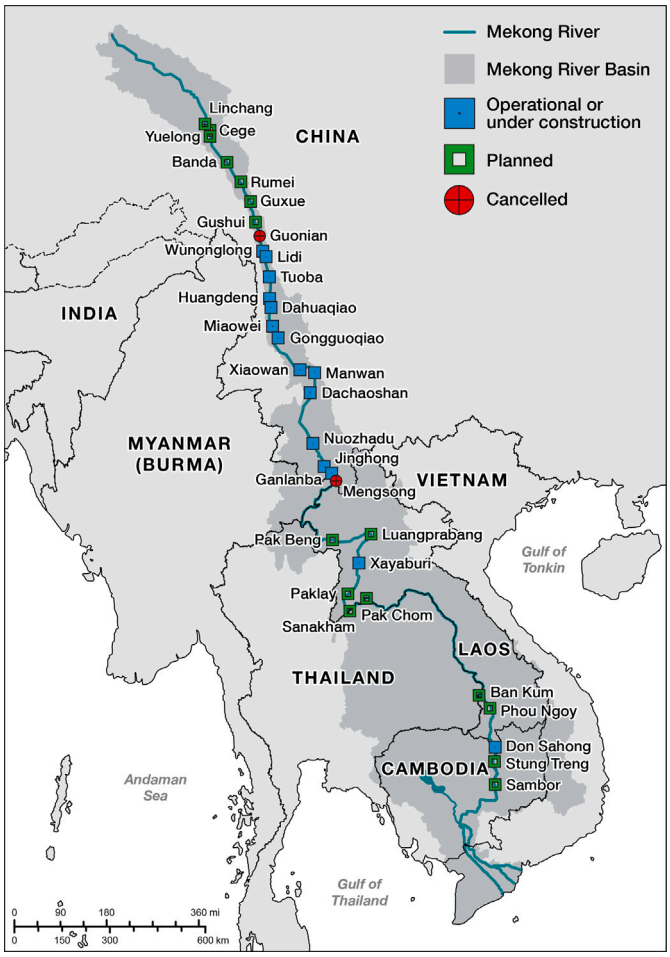
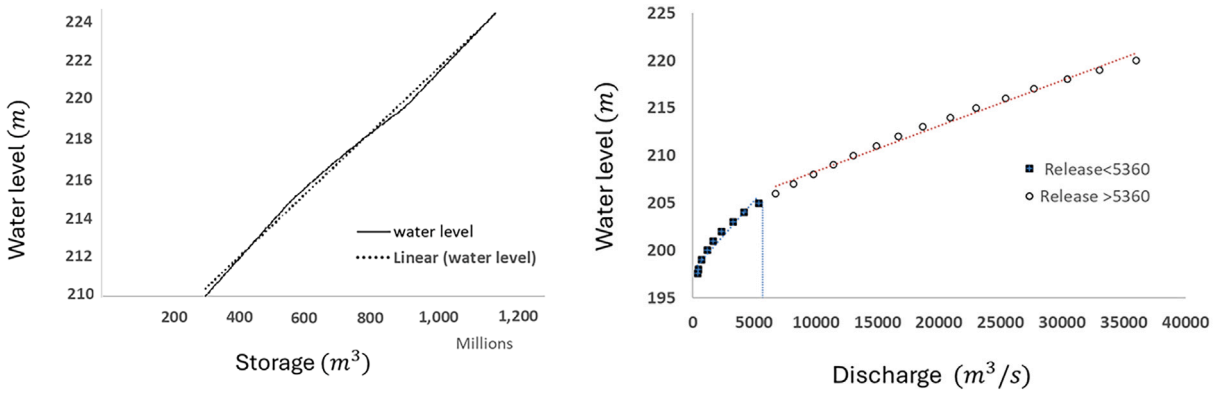


Fig. 4. Map of dams along the Mekong mainstream.



(a) Storage-water level curve (MRC, 2018) (b) Rating curve (MRC, 2018)

Fig. 5. Storage and discharge vs. water level.

planned, 2 decommissioned, and 7 canceled. These include 608 hydropower and 447 non-hydropower dams, the latter supporting water supply, flood control, and irrigation [7]. Currently, 15 hydropower dams operate along the river's mainstream, with 9 more planned [7].

The Sanakham project will be the fifth hydropower station in the Mekong River cascade in Laos, located between Vientiane and Xayabury (Fig. 4). It is a low-head, high-flow RoR facility with daily regulation capability. Construction began in 2020 and is scheduled for completion in 2028 [47]. The plant has an installed capacity of 684 MW, supported by a reservoir with 827 million m^3 of storage and a normal water level of 220 meters. It consists of 12 identical 57 MW bulb turbines housed in a single structure on the main river channel, operating as a RoR facility with a maximum discharge capacity of 5800 m^3/s. The head ranges from 4 to 20 meters. The plant shuts down for safety when the head drops below 4 meters. We model Sanakham's operations as a single aggregate unit in our case study. We assume a turbine efficiency of 95 % and a generator efficiency of 90 %, resulting in a total efficiency of $e = 85.5$ %. The gravitational constant is 9.81, m/s^2, and the constant related to water density and unit conversion is set as $a_t = 10^{-3}$. The coefficient for head loss in the net head calculation is assumed to be $C_L = 4.56 \times 10^{-14}$.

The relationship between upstream water level and storage volume is illustrated in Fig. 5a [47]. The relationship between upstream water level and storage volume is based on the reservoir's topography. Because the minimum water level in this reservoir is 210 m, the relevant segment of the curve in Fig. 5a can be approximated with a linear function (dotted line). This linear function represents $\hat{f}(s_t)$ in Constraints (18), specifying that the upstream water level $\hat{H}_t$ is a function of the storage volume s_t at time t. For the initial conditions, the storage level is assumed to be at 218 m with its corresponding storage volume given by Fig. 5a.

We applied our model for daily operations with a maximum storage change of 5 % within each 5-day time window and smoothing factors $\psi = 0.2$ and $\psi' = 0.3$ in Constraints (13) and (14). Under normal operations, the storage volume is not allowed to be less than 760 m^3, or equivalently, the water level cannot drop below 218 m. Table 3 provides the maximum allowable water release values corresponding to various upstream water level conditions [47]. These values serve as thresholds used in Constraints (9) and (10).

When flow is regulated, the downstream water level can be estimated using a rating curve specific to the dam site. Fig. 5b illustrates the projected rating curve for the Sanakham project [47]. The rating curve shows the relationship between the discharge and the downstream water level. We approximate the rating curve using a piece-wise linear function. If the discharge is less than 5360 m^3/s, the slope and y-intercept are set to 0.0005 and 203.56, respectively. Otherwise, they are 0.0015 and 197.87, respectively. The threshold value is determined based on the quality of the estimation of the resulting segments. These linear functions represents $\check{f}(Q_t)$ in Constraints (19), specifying that the downstream water level $\check{H}_t$ is a function of the water release Q_t at time t. The water inflow data (i.e., I-parameters) is based on the release from the nearest upstream gauge, Luang Prabang, and adjusted using external annual precipitation (i.e., E-parameters) [47].

In the following sections, we evaluate the performance of our proposed HPF approximation, linearization method, and operational curve. To ensure robust assessment, we distinguish between in-sample and out-of-sample evaluations. While the hydropower production function and operational curves were estimated using data-driven methods, validating their performance on independent test data is essential. We detail the sources of both training and test datasets in the sections that follow.

Table 3
Calculation results of water release capacity.

Upstream water level (m)	Maximum water release (m^3/s)
210.05	12,303
211.20	14,429
212.38	16,018
212.90	16,261
213.85	17,921
214.50	19,061
215.58	21,252
216.50	23,185
216.95	24,016
217.75	25,542
218.66	27,248
220.78	31,755
220.98	33,043
221.70	34,796
223.30	38,967

6.1. Analyzing the performance of the proposed HPF approximation

We use the operational parameters of the Sanakham plant to validate the performance of the HPF from Section 3. Since the water head range is between 4 m and 20 m, we find $\beta_j = f(\hat{h}_j)$ for $j \in \{4, \ldots, 20\}$ as the solution to a least-squares problem between the exact function in (1) and the approximation in (4). To determine each β-correction factor, we sample multiple values of storage volume and discharge level for each head value, also obtaining the corresponding upstream and downstream water levels from the storage-water level and rating curves. The 2-step procedure is described as follows:

Step 1: Calculate Eq. (1) using sampled values of discharge between m_t and $q_{|K|}$, storage volume between s_t^l and s^c, and upstream and downstream water levels within the bounds specified by the storage-water level and rating curves.

Table 4
Correction coefficient estimates.

Water head (m)	Coefficient	Water head (m)	Coefficient	Water head (m)	Coefficient
4	0.09	10	0.43	16	0.77
5	0.15	11	0.49	17	0.83
6	0.21	12	0.55	18	0.89
7	0.26	13	0.60	19	0.94
8	0.32	14	0.66	20	1
9	0.38	15	0.72		

Step 2: For each gross head value $\hat{h}_j = \{\hat{h}_1, \ldots, \hat{h}_{|K|}\}$, set $\lambda_j = 1$, and solve a least-squares problem to determine $f(\hat{h}_j)$ by minimizing the error between the hydropower production obtained from Eq. (1) in Step 1 and the values from Eq. (4). Set $\beta_j = f(\hat{h}_j)$.

Table 4 shows the correction coefficients after solving the least-squares problems for a water head range with $\hat{h}_0 = 4$, $\hat{h}_{|K|} = 20$, and 1 m increments. The correction factor equals 1 when the water head is at its maximum value and approaches zero when the water head is at its minimum. This result is expected, as the HPF approximation in (4) is based on the maximum head before applying the correction. To maintain a reduced number of decision variables, we use a subset of water head intervals with breakpoints $\hat{h}_0 = 0, \hat{h}_1 = 5, \hat{h}_2 = 10, \hat{h}_3 = 15, \hat{h}_4 = 20$ in the optimization model. This two-step procedure is not limited to Eq. (1). Samples can be evaluated using any benchmark HPF, including those that model efficiency more explicitly or account for power production from a subset of operating units. For example, an efficiency term $\eta(x)$ can be introduced as a function of release. For each sampled head level $\hat{h}_j$ and turbined outflow x in Step 1, the benchmark power output can incorporate variable efficiency $\eta(x)$. Then, the correction factor $f(\hat{h}_j)$ estimated in Step 2 adjusts Eq. (4) to align with the revised benchmark, capturing the variation in efficiency.

To validate the accuracy of our model, we use an out-of-sample test dataset, consisting of $n = 98$ pairs of water release and storage values, which we use to infer water levels. This dataset is used to calculate values for Eq. (1), as well as for the HPF in Eq. (4). Table 5 reports the absolute percentage error in hydropower production between Eqs. (1) and (4), calculated relative to Eq. (1). We use the results from Eq. (1) as a benchmark in the absence of real hydropower production data. The columns represent the sampled total release as a percentage of the maximum discharge, while the rows indicate the sampled storage volume as a percentage of the storage capacity. Note that the columns include values greater than 100 % to account for spillage and its impact on the downstream water level and head.

The results in Table 5 show that the proposed HPF is an accurate approximation of the HPF in (1), particularly when the reservoir operates as a RoR system with storage volume greater than 80 % of its capacity. In this case, the error is less than 1 %. The maximum and average overall errors are 3.7 % and 0.95 %, respectively. The outputs marked as (-) indicate that the corresponding combination of storage volume and total release falls outside the acceptable range of gross head values, meaning the gross head exceeds the maximum allowable value of 20 m.

6.2. Analyzing the performance of the linearized HPF

In this section, we assess the quality of the first-order approximation approach in (36) with respect to (20). To obtain the sampling points $\left(\overline{x}^{\omega}, \overline{\lambda}^{\omega}\right)$ to construct (36), we first draw 16 values from the water head intervals $(0, \hat{h}_1]$, $(\hat{h}_1, \hat{h}_2]$, $(\hat{h}_2, \hat{h}_3]$, and $(\hat{h}_3, \hat{h}_4]$, as described in Section 6.1, with four sampling values in each interval. These values enable us to capture the system's behavior across different water head levels with sufficient granularity. Within each interval, we also consider discharge values evenly distributed between one-quarter of the maximum discharge and the maximum value. This produces enough combinations of discharge levels and head values. We then set the $\overline{\lambda}^{\omega}$-values for each sample point equal to either 0 or 1, according to their water head layer. We construct the inequalities in (36) using the β-parameters from Section 6.1 and solve the optimization model with them. Table 6 shows the absolute percentage error between the first-order approximation in (36) and Eq. (20), calculated relative to (20), for $n = 60$ combinations of initial storage volume and required release. The columns represent the total release as a percentage of the maximum discharge, and the rows represent the storage volume as a percentage of the storage capacity. The results demonstrate that the proposed first-order linear approximation

Table 5
Absolute percentage error between Eqs. (1) and (4).

Storage volume (% of maximum storage)	Total release (% of maximum discharge)													
	30 %	40 %	50 %	60 %	70 %	80 %	90 %	100 %	120 %	140 %	170 %	200 %	260 %	300 %
50 %	2.92	3.60	3.70	3.50	3.40	2.70	1.34	0.58	0.65	0.89	1.13	1.54	2.33	3.6
60 %	1.74	2.36	2.55	2.44	2.41	1.98	0.99	0.40	0.49	0.61	0.79	1.03	1.81	2.19
70 %	0.62	1.20	1.43	1.43	1.50	1.28	0.64	0.26	0.31	0.41	0.55	0.65	1.13	1.54
80 %	0	0.57	0.83	0.90	1.04	0.92	0.46	0.21	0.24	0.31	0.41	0.55	0.89	1.13
90 %	–	0	0.29	0.43	0.63	0.62	0.32	0.15	0.20	0.24	0.35	0.41	0.65	0.89
95 %	–	–	–	–	–	0.11	0.08	0.05	0.08	0.14	0.20	0.24	0.41	0.55
100 %	–	–	–	–	–	–	–	0.01	0.03	0.08	0.14	0.20	0.31	0.41

Table 6
Absolute percentage error between first-order approximation in (36) and (20).

Storage volume (% of maximum storage)	Total release (% of maximum discharge)										
	40 %	50 %	60 %	70 %	80 %	90 %	100 %	120 %	150 %	170 %	200 %
60 %	1.51	0.51	1.21	2.49	0.95	1.97	1.70	1.69	1.68	1.65	1.65
70 %	2.60	0.33	3.45	1.82	1.13	0	1.53	1.47	1.44	1.41	1.41
80 %	1.27	1.09	2.18	1.23	1.29	0	1.37	1.26	1.25	1.24	1.24
90 %	2.26	0.53	2.27	1.67	1.17	0	1.49	1.38	1.42	1.12	1.11
100 %	–	–	–	–	–	–	1.34	1.25	1.23	1.20	1.20

provides an accurate and consistent overestimation of the HPF in (20) when the model is solved to optimality.

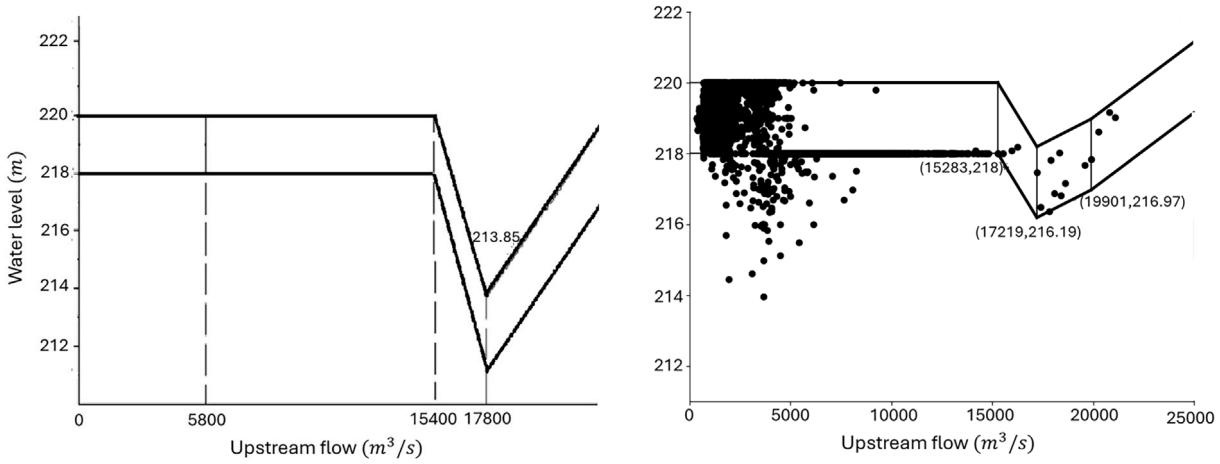
6.3. Analyzing the performance of the operational curves

We also compare the results of our proposed model with a modified version that incorporates an operational curve. Two types of curves are used: a data-driven operational curve derived from optimal plans using (42)–(51), and a technical curve based on engineering design for Sanakham [47]. The technical curve, grounded in hydrological calculations, prioritizes flood control, while the data-driven curve reflects the tradeoff between flood control and hydropower production. The technical curve for Sanakham was developed through hydrologic modeling and simulation across various inflow scenarios. The goal is to evaluate hydropower output when decisions are guided by these curves. Technical operational curves are widely used in water resource management to link upstream inflows to reservoir levels, primarily serving as a tool for real-time decision-making based on flow monitoring. Fig. 6 displays both curves.

The technical operational curve in Fig. 6a has the following features:

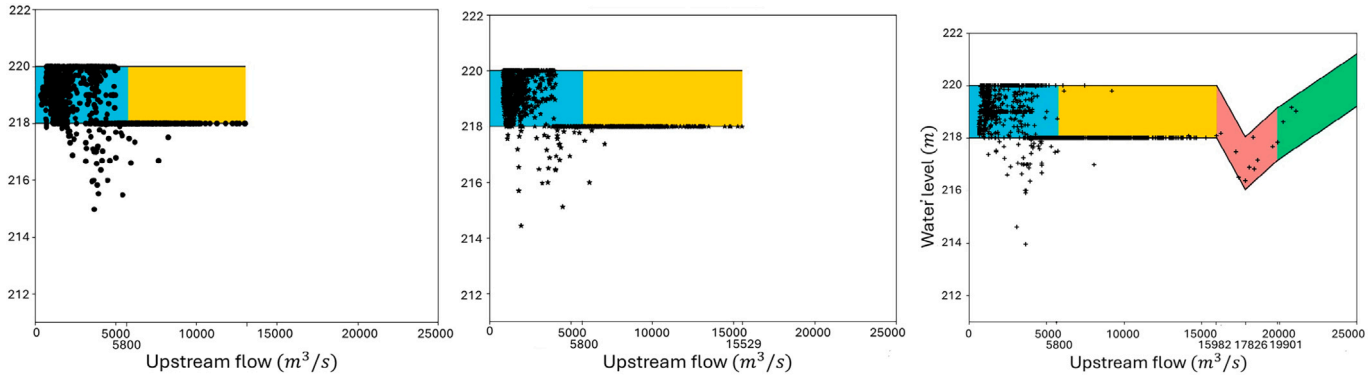
- The first breakpoint occurs at 5800 m^3/s, the maximum inflow for full-capacity turbine operation. Below this, water level should range between 218–220 m.
- Between 5800 and 15,400 m^3/s, turbines operate at full capacity and excess inflow is spilled. The water level is maintained at 218 m with a 2 m buffer.
- Between 15,400 and 17,800 m^3/s, reservoir releases are adjusted to create capacity for expected floods, balancing flood control and future water demands. Inflows above 17,800 m^3/s outpace the maximum total release, resulting in an increase of the reservoir level.
- Inflows are lagged by two days (from Luang Prabang to Sanakham) to allow operational response. For every 1200 m^3/s excess inflow, 0.1 billion m^3 storage is required. The system can release 2400 m^3/s per day, gradually lowering water levels in anticipation of flood arrival. Due to the RoR nature of the reservoir, no more than 0.1 billion m^3 of water is allowed to be released each day, which is equivalent to 2400 m^3/s for 2 days.

The data-driven operational curve in Fig. 6(b) is derived using the model from Section 5. The training data consist of the optimal solutions (daily) of the model for 10 years from 1999 to 2010, excluding 2008, which is reserved for out-of-sample testing. It is also possible to train operational curves for different climate conditions. Fig. 7 depicts data-driven operational curves that are designed to capture different precipitation scenarios, such as moderate, wet, and dry years. Moderate years are defined as those with typical wet seasons, wet years experience exceptionally strong wet seasons, and dry years undergo poor wet



(a) Technical operational curve (b) Data-driven operational curve

Fig. 6. Operational curves for Sanakhan.



(a) Dry years (b) Moderate years (c) Wet years

Fig. 7. Operational curves for three hydrological conditions.

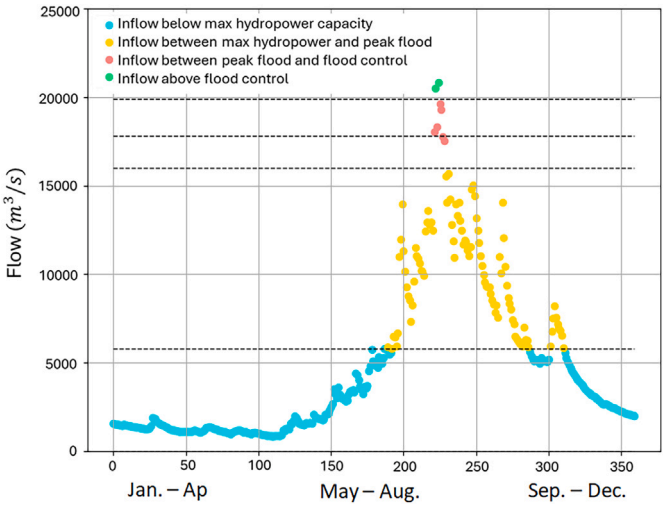


Fig. 8. Hydrology for the test year.

seasons. Using these operational curves, operators can tailor hydropower production plans according to expected hydrological conditions.

We use input data from 2008 and solve the model to prescribe daily operations. We did not use 2008 in the training of the data-driven operation curve or the estimation of HPF parameters in (36). Figs. 8–12 summarize the input data and results after solving the optimization model. These include the hydrology of the test year (Fig. 8), the technical and data-driven operational curves (Fig. 9), and the operational results (Figs. 10 and 13). The operational results include upstream and downstream water levels (Fig. 10), discharge for hydropower (Fig. 11), spillage (Fig. 12), and hydropower production (Fig. 13).

Fig. 8 shows that for some days, the reservoir's inflow rate is greater than the peak flood release (15,982 m^3/s) (around day 230), which must be handled via additional spillage. In Figs. 10 and 13, Part (a) shows the results from the base model in (5)–(34) with (20) replaced by (36). Part (b) shows the results after embedding the technical operational curve as a constraint to limit the water level based on the lagged inflow, while Part (c) shows the results after embedding the data-driven operational curve. Fig. 10 illustrates the downstream and upstream water levels and their difference over time. The color of the dots in the upstream water level capture the range in the inflow rate as in Fig. 8. Figs. 11 and 12 show the discharge for hydropower and spillage, respectively, whereas Fig. 13 shows the daily hydropower production. In Fig. 13, the solid line represents the results using the first-order approximation, while the dashed line depicts the hydropower production using (20). We see that the first-order approximation consistently overestimates the hydropower production for the same input, with a maximum error of 3.45 %. Figs. 11(a) and 13(a) show that the water released for hydropower and the hydropower produced are not at full capacity until day 180, where the inflow surpasses 5800 m^3/s. The water release for hydropower production remain at full capacity until the inflow returns to 5800 m^3/s in day 270. Fig. 12(a) shows that there is spillage only if the channel is at maximum capacity.

Figs. 10–13 (Part b) depict the model results after enforcing the technical operational curve. Fig. 10(b) shows a smoother upstream water level compared to Fig. 10(a), which reduces the hydropower production by only 2 %, as shown in Figs. 13(a) and (b). Fig. 13(b) exhibits a sudden drop in the hydropower produced in day 230. This is due to

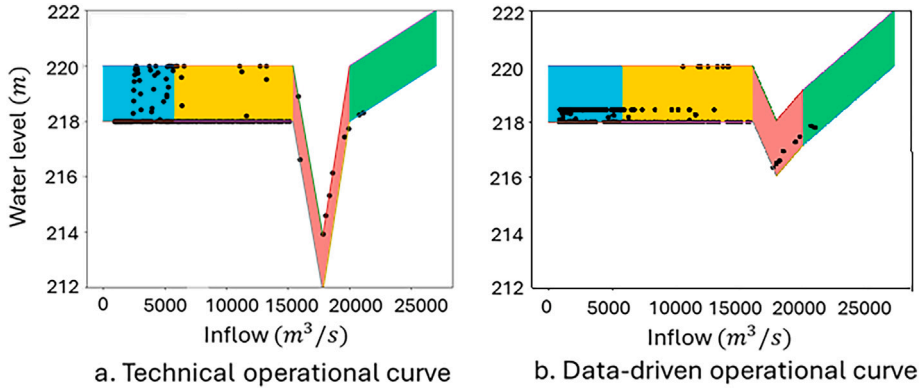


Fig. 9. Operational curves and operational decisions satisfying the curves.

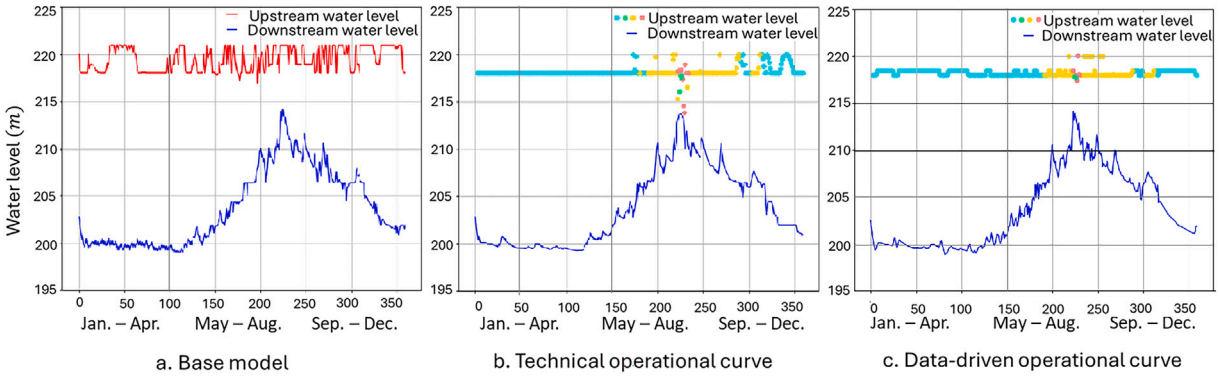


Fig. 10. Upstream and downstream water levels.

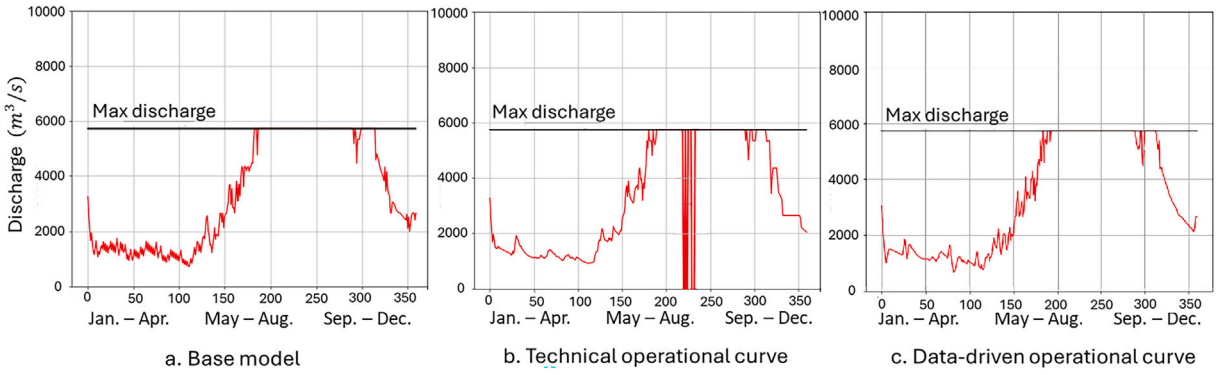


Fig. 11. Discharge for hydropower.

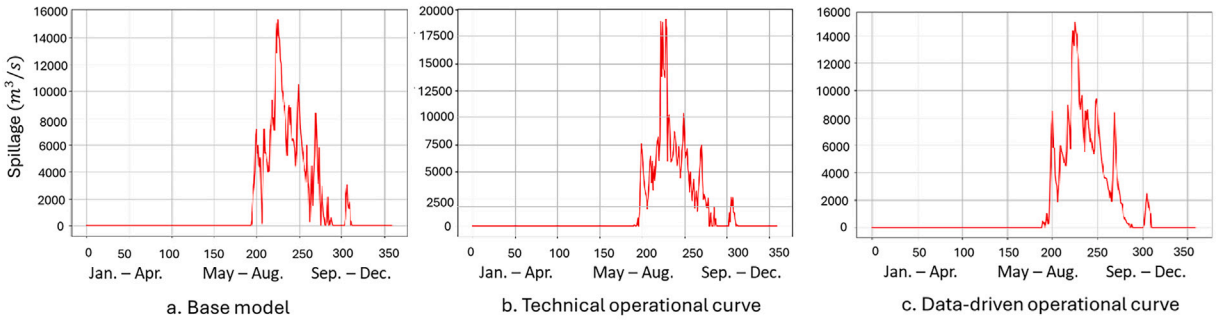


Fig. 12. Spillage.

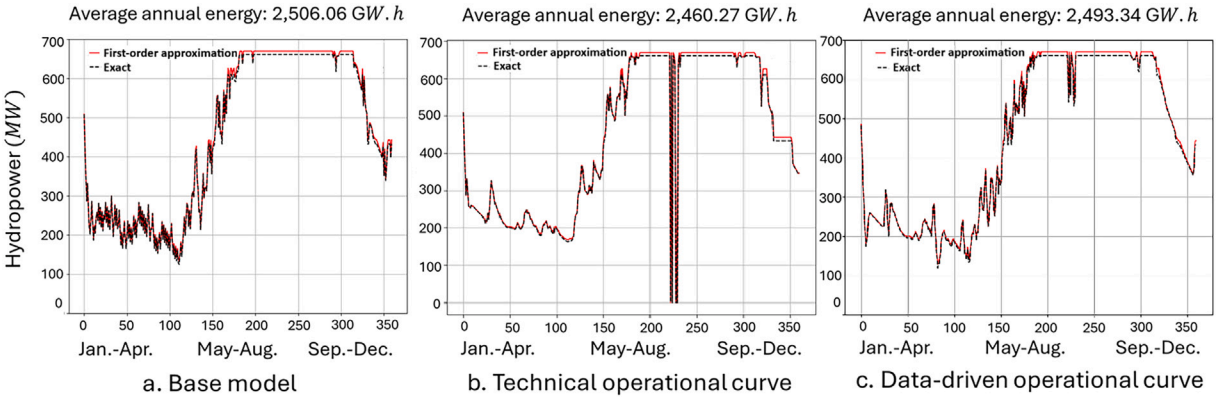


Fig. 13. Hydropower production.

the constraint enforcing the powerhouse shut-down when the difference between upstream and downstream water levels is less than 4 m. This can be observed in Fig. 10(b).

Figs. 10–13 (Part c) depict the model results after incorporating the data-driven operational curve. Figs. 9(a) and 10(b) show that during the flood seasons, generally between May and August, the upstream water level decreases to 213.85 m two days before the flow increases to free the necessary storage to absorb the additional water. The same behavior is observed in Figs. 9(b) and 10(c) where the upstream water level decreases to 216.04 m two days before the flow increases. Before the wet season, the water level remains between 218 m and 220 m with fewer fluctuations compared to Fig. 10(a). Fig. 13(c) shows that the annual average hydropower produced is 2493.34 GW, which is closer to 2506.06 GW obtained by the base model compared to the total hydropower from the model enforcing the technical operational curve. Fig. 14 illustrates the water level after embedding the data-driven operational curve in comparison to the base model. This demonstrates that using the operational curve reduces the volatility in the water level, which is desirable in real operations.

Table 7 compares the performance of our model with and without operational curves for daily operations from 1999 to 2020. The models with operational curves require water releases to meet water level requirements specified by the technical and data-driven curves, while the model without an operational curve (base model) is free to set water releases as long as other technical requirements are satisfied (e.g., maximum turbine discharge). Fig. 9 illustrates how the operational plans

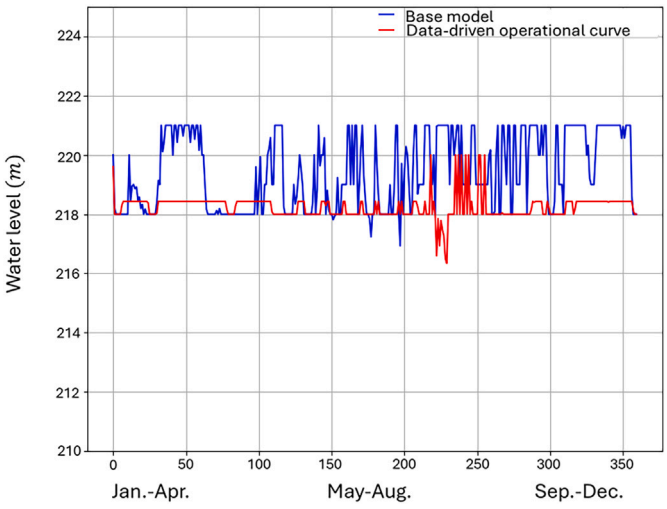


Fig. 14. Upstream water levels before and after embedding operational curve.

adhere to each operational curve. For these experiments, we used a desktop computer with an Intel Core i7 1.10 GHz processor and 16.0 GB RAM, using Python integrated with CPLEX 12.8 as the optimization solver.

Operational curves can be perceived as rigid or overly conservative policies that may not fully account for the dynamic nature of water systems and operational needs. However, the buffer ρ provides flexibility for operational adjustments, helping to mitigate this rigidity. We observe that enforcing the operational curve results in minimal hydropower loss compared to the base model. In most years, the technical operational curve leads to greater losses than the data-driven curve. This outcome is expected, as the data-driven curve was trained on optimal plans that balance hydropower production and flood control, whereas the technical operational curve is more conservative, focusing solely on flood control and occasionally releasing more water than necessary. This demonstrates that operators must select not only the appropriate operational curve for anticipated conditions (dry, moderate, or wet) but also a curve that aligns with their tolerance for unexpected future conditions.

The model requires calibration when incorporating the technical and data-driven operational curves in order to match the input hydrology. For instance, the model for 2004 becomes infeasible due to inflow fluctuations and the smooth release in Constraints (13). However, the operational curve models regain feasibility by relaxing the smoothing factor–ψ–from 0.2 to 0.3 to allow more drastic changes in the release. To evaluate the robustness of our operational curves under extreme hydrological conditions, we conducted sensitivity analyses for three scenarios: extreme drought, extreme rainfall, and extreme inflow volatility. The results are presented in Appendix B.

7. Conclusion and future research

This study focuses on optimizing reservoir operations to maximize hydropower production. The model used in this research addresses long-term challenges involving significant changes in net head, a crucial factor for accurate hydropower calculations. To achieve realistic results, the model incorporates key features such as improved net head calculations and a head-dependent hydropower function, which is approximated for computational efficiency.

The introduction of two types of operational curves plays a vital role in controlling fluctuations in water flow. This work emphasizes the importance of optimizing operational curves to manage the impacts of climate change and input flow fluctuations on hydropower production. Although more restrictive than the base model, the results demonstrate that the model with a data-driven operational curve performs closer to the optimal solution compared to the model with a technically-based operational curve. This suggests that incorporating data-driven approaches in operational curve optimization can yield better results and adapt more effectively to changing conditions. The main challenge in implementing data-driven operational curves lies in the training process, which requires access to historical hydropower production plans or a reliable method for estimating them.

It is also important to consider the broader context of this research. The study emphasizes the significance of alternative and renewable energy sources, particularly focusing on hydropower operations. However, a more comprehensive analysis is necessary to evaluate the suitable locations and feasibility of hydropower projects. Factors such as environmental impact, social considerations, and economic viability should be considered to make informed decisions about the development and deployment of hydropower facilities.

The presented case study focuses on the operation of Sanakham, a dam project in the Mekong River Basin. For context, governments of the Lower Mekong and Upper Lancang are constructing numerous dams (see Fig. 5a), many of which are mainstem RoR structures. A common assumption with mainstem RoR dams is that they have “no storage” capacity and thus pass water through with minimal impact on downstream flow regimes. We challenged this assumption and demonstrated that although RoR dams have significantly less storage than traditional reservoirs, the storage available in many mainstem RoR structures on the Mekong can be optimized for greater energy production.

A possible future direction is optimizing hydropower operations across a network of reservoirs while minimizing negative environmental impacts. This aspect is not addressed in this paper due to the extensive data required to incorporate these impacts into the model. Future research could explore an optimization model that balances competing objectives, such as maximizing hydropower production

Table 7
Total hydropower and solution time with and without operational curve.

Scenario	Total hydropower (GW)					Run time (s)		
	Base model	Technical OC	Data-driven OC	% Loss in technical OC	% Loss in data-driven OC	Base model	Technical OC	Data-driven OC
2010	1845.13	1843.13	1843.13	0.10	0.10	35	22	22
2008	2506.06	2460.27	2493.34	1.82	0.51	45	30	31
2007	2300.09	2297.06	2297.57	0.13	0.11	47	31	27
2006	2299.67	2297.65	2297.65	0.09	0.09	34	29	27
2005	2344.52	2338.28	2338.28	0.27	0.27	36	29	27
2004*	2417.17	2412.06	2414.21	0.21	0.12	44	39	39
2003	2302.89	2300.44	2300.44	0.11	0.11	46	32	27
2002	2533.41	2485.50	2519.48	1.89	0.55	34	31	30
2001	2577.19	2571	2572.89	0.24	0.17	35	28	27
2000	2470.38	2468.10	2467.23	0.09	0.13	37	35	29
1999	2265.32	2259.19	2260.84	0.27	0.19	45	35	31

while considering river flow, flood levels, and food and agricultural outcomes. Integrating unit commitment decisions and turbine-specific efficiency into the model can enhance model accuracy, but these improvements substantially increase computational complexity, presenting a trade-off between accuracy and tractability that merits further study.

CRediT authorship contribution statement

Fatemeh Karamyar: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Jorge A. Sefair:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yuhang Wei:** Writing – original draft, Validation, Software, Investigation, Data curation. **John L. Sabo:** Writing – original draft, Supervision, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Jorge A. Sefair reports financial support was provided by National Science Foundation. John L. Sabo reports financial support was provided by National Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Model size

To describe the computational complexity of the proposed MILP formulation (base model) for hydropower optimization, we summarize the total number of binary variables and constraints (excluding (30)–(34)) as a function of the model parameters. Table A.8 provides a parametric expression for the model size, where $|T|$ is the number of time periods, $|J|$ is the number of discretized head intervals, $|K|$ is the number of upstream water level thresholds, and Ω is the number of samples in the first-order HPF approximation.

Table A.9 summarizes the model size for the operational curve model. The size of the model depends on the number of input points ($|C|$) defining the curve and the number of piecewise linear segments ($|\mathcal{P}|$) used to approximate the operational behavior.

Table A.8
Parametric size of the hydropower model.

Model component	Number
Binary variables (total)	$\|T\|(\|J\|+\|K\|+2)$
α_{jt} (head interval indicator)	$\|T\|\|J\|$
y_{tk} (water level threshold indicator)	$\|T\|\|K\|$
d_t (head difference indicator)	$\|T\|$
θ_t (turbine max flow indicator)	$\|T\|$
Constraints (total)	$\|T\|(\|J\|+2\|K\|+\|\Omega\|+23)$

Table A.9
Parametric size of the operational curve model.

Model component	Number
Binary variables (total)	$2\|C\|\|\mathcal{P}\|$
δ_{cp} (point-to-segment assignment)	$\|C\|\|\mathcal{P}\|$
σ_{cp} (segment breakpoints)	$\|C\|\|\mathcal{P}\|$
Constraints (total)	$7\|C\|\|\mathcal{P}\|-3\|C\|+3(\|\mathcal{P}\|-1)$

Appendix B. Sensitivity analysis

We consider the following additional input scenarios.

- **Extreme drought scenario:** A synthetic dry year in which inflows remain below turbine capacity throughout the year, including during the wet season (see Fig. B.15(a)). Figs. B.15(b–c) compare water level versus inflow behavior across the base model and models using technical and data-driven operational curves. While the base model shows greater fluctuations, both operational curve models maintain water levels within a narrower range, indicating improved stability. Total hydropower generation for the base model was 1293.44 GWh, compared to 1291.32 GWh for both operational curve models, a negligible difference of only 0.16 %, demonstrating minimal energy loss under extremely dry conditions.
- **Extreme rainfall scenario:** A synthetic wet year with inflows consistently above turbine capacity and extreme peaks exceeding the flood control threshold (Fig. B.16(a)). As shown in Fig. B.16(b, c), the data-driven operational curve stabilizes water levels within a narrower range, while the base model shows greater fluctuations. The technical operational curve initially led to infeasibility, so we relaxed the allowable storage reduction to 25 % (up from 8 %, reflecting the RoR nature of the dam) to ensure feasibility. In this scenario, the base model produced 3574.86 GWh, the data-driven curve yielded 3543.83 GWh (a 0.87 % reduction), and the technical curve produced 3399.87 GWh under the relaxed conditions.
- **Extreme inflow volatility scenario:** A synthetic year with inflows up to three times more volatile than the most variable historical year (Fig. B.17a–c). To maintain feasibility, we relaxed the allowable storage reduction to 30 % (up from 8 %) and increased the smoothing limit on total release to 35 %. Despite these adjustments, the technical operational curve model remained infeasible. The base model generated 2581.3 GWh, while the data-driven operational curve model produced 2567.2 GWh—a modest 0.55 % reduction.

Figs. B.18–B.31 illustrate the water level, discharge, spillage, hydropower production, and inflows and outflows for each scenario. Figs. B.21 and B.26 show that while operational curves primarily regulate upstream water levels, they also reduce outflow fluctuations indirectly. Unlike the base model, which follows a purely hydropower-maximizing strategy, the operational curves promote discharge stability and help maintain environmental flow continuity with minimal impact on hydropower production. One key insight from Fig. B.31, under the extreme volatility scenario, is that the outflow becomes less volatile than the inflow due to the model's smoothing constraints, which supports more stable reservoir operations.

A key finding is the infeasibility of the models using the technical operational curve under extreme rainfall and highly volatile inflow scenarios. These conditions pose greater operational challenges due to the need to accommodate large or drastically different inflows from previous periods. In contrast, the data-driven operational curve models remained feasible and produced near-optimal plans with minimal loss in hydropower generation compared to the base model. However, in some cases they triggered more frequent turbine shutdowns due to stricter limits on maximum water levels (and thus head) imposed by the data-driven curve. These results highlight the importance of (re)training operational

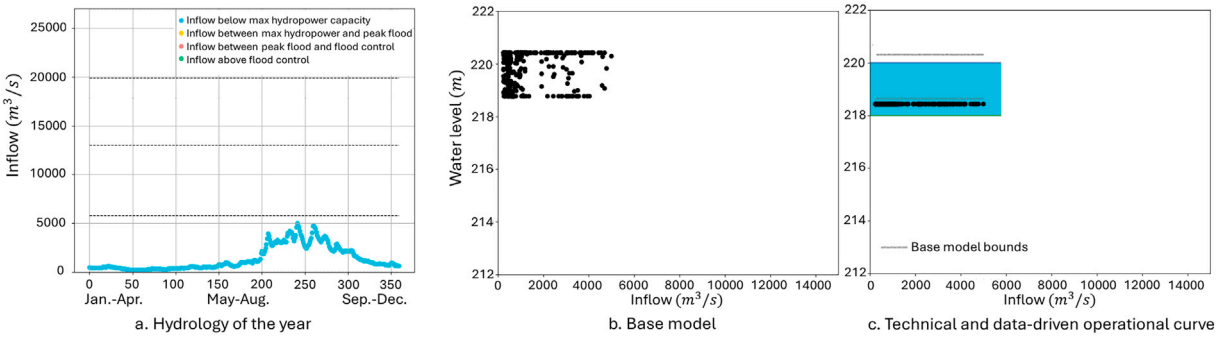


Fig. B.15. Operational decisions for extreme drought.

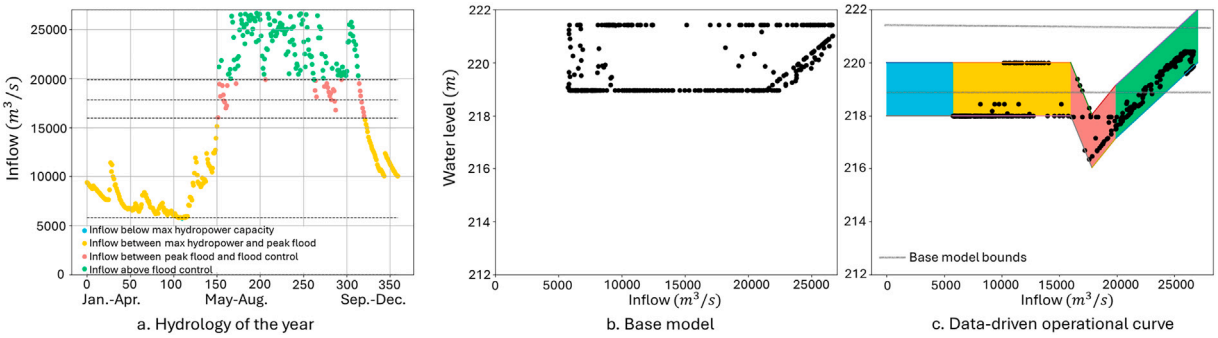


Fig. B.16. Operational decisions for extreme rainfall.

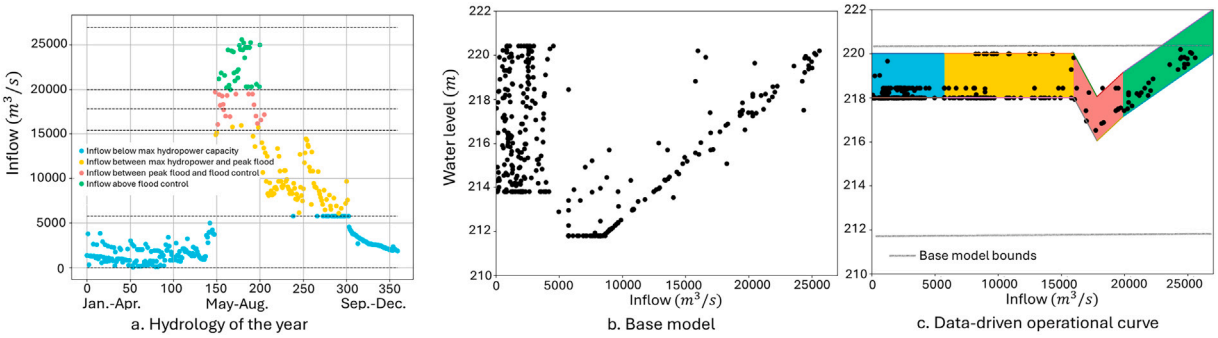
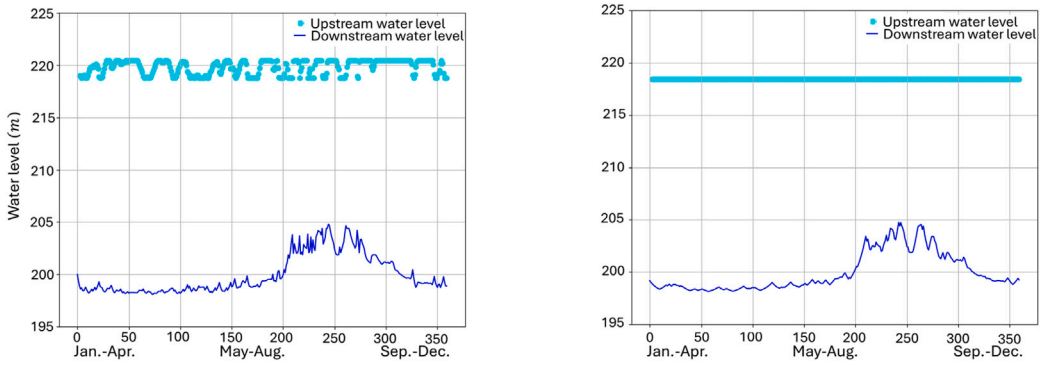
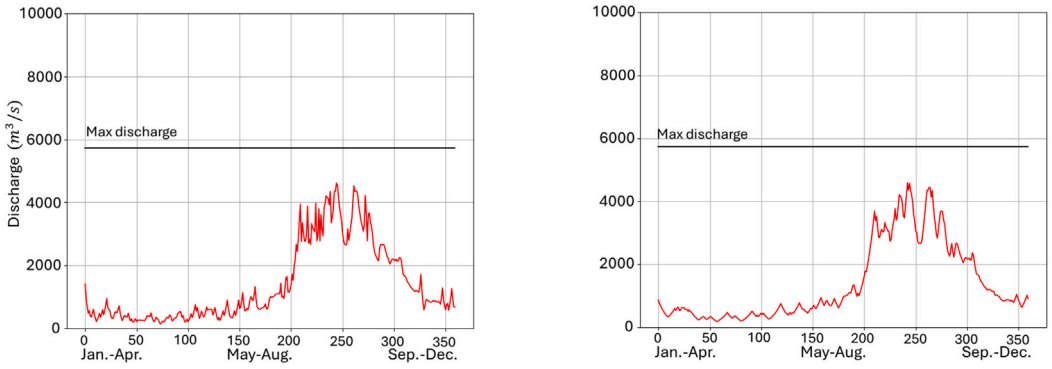


Fig. B.17. Operational decisions for extreme inflow volatility.



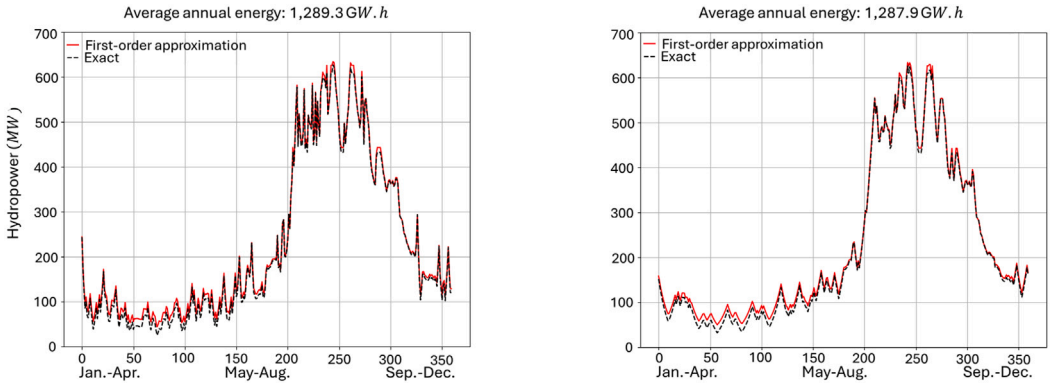
(a) Base model (b) Technical and data-driven operational curve

Fig. B.18. Upstream and downstream water levels (extreme drought).



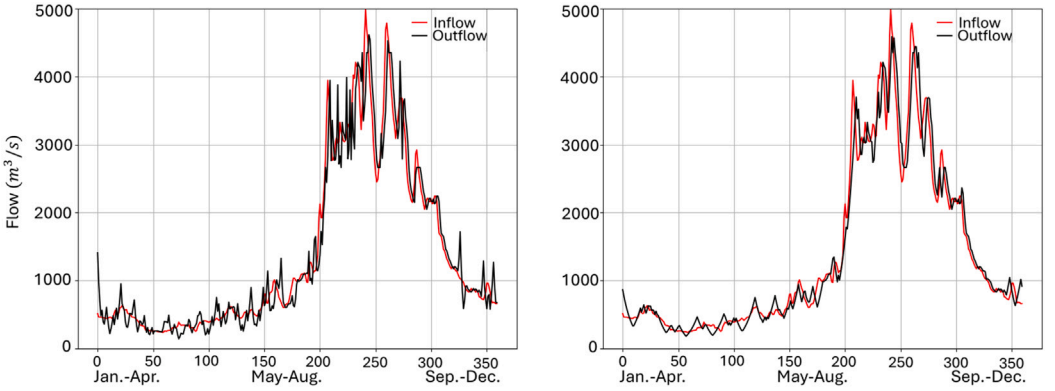
(a) Base model (b) Technical and data-driven operational curve

Fig. B.19. Discharge for hydropower (extreme drought).



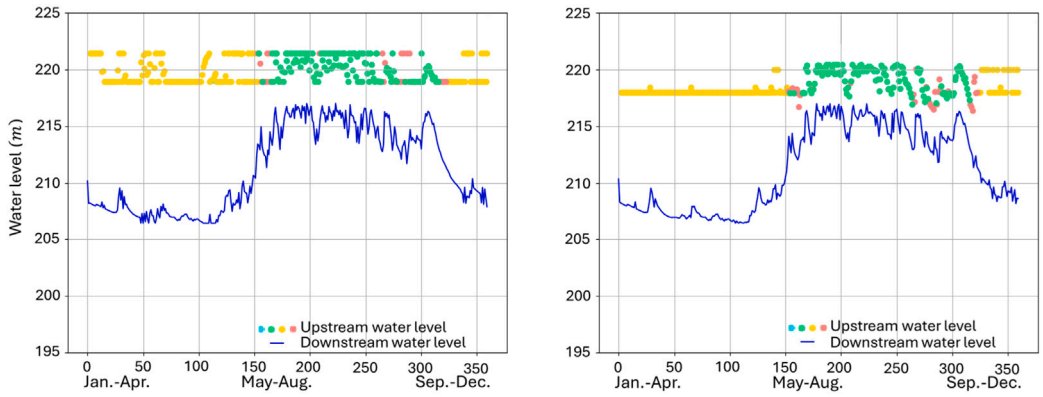
(a) Base model (b) Technical and data-driven operational curve

Fig. B.20. Hydropower production (extreme drought).



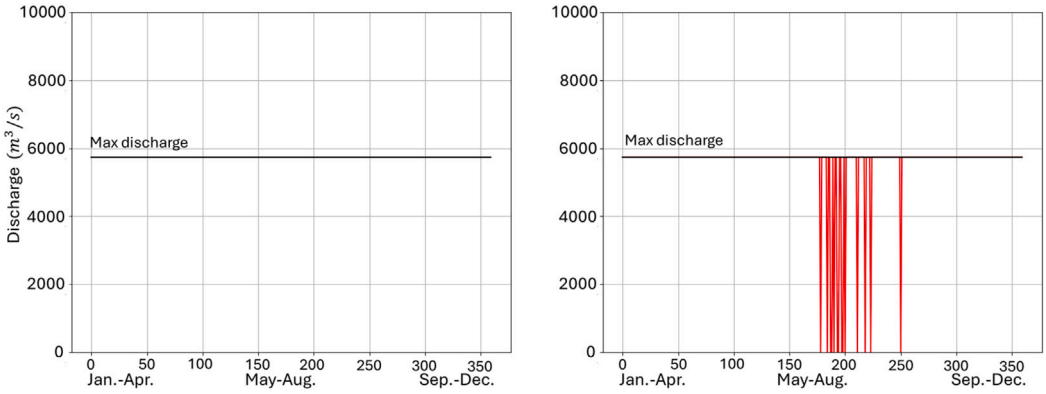
(a) Base model (b) Technical and data driven operational curve

Fig. B.21. Inflow and total outflow (extreme drought).



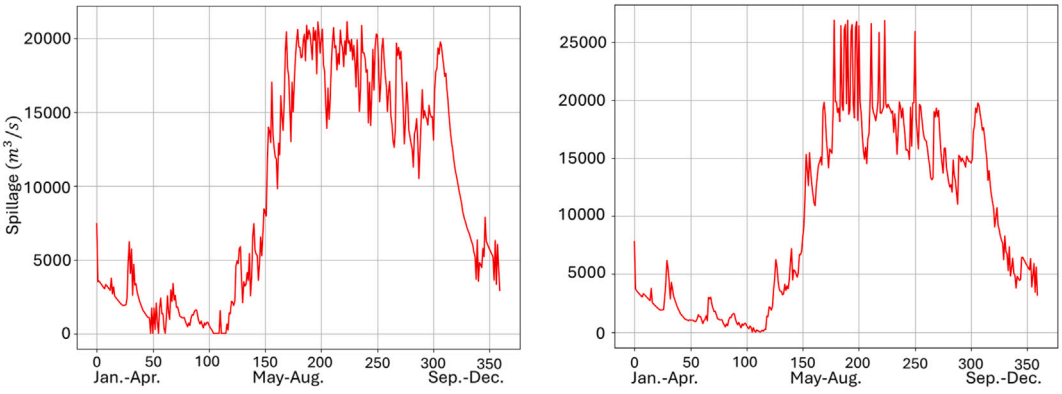
(a) Base model (b) Data-driven operational curve

Fig. B.22. Upstream and downstream water levels (extreme rainfall).



(a) Base model (b) Data-driven operational curve

Fig. B.23. Discharge for hydropower (extreme rainfall).



(a) Base model (b) Data-driven operational curve

Fig. B.24. Spillage (extreme rainfall).

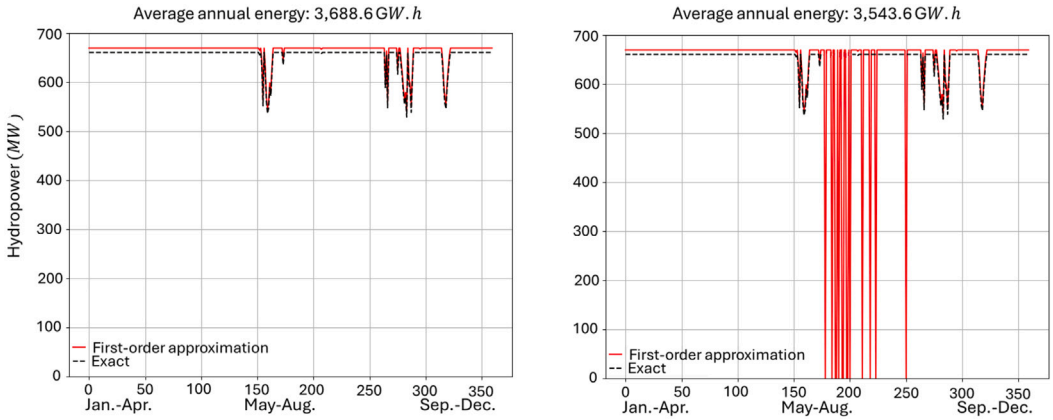


Fig. B.25. Hydropower production (extreme rainfall).

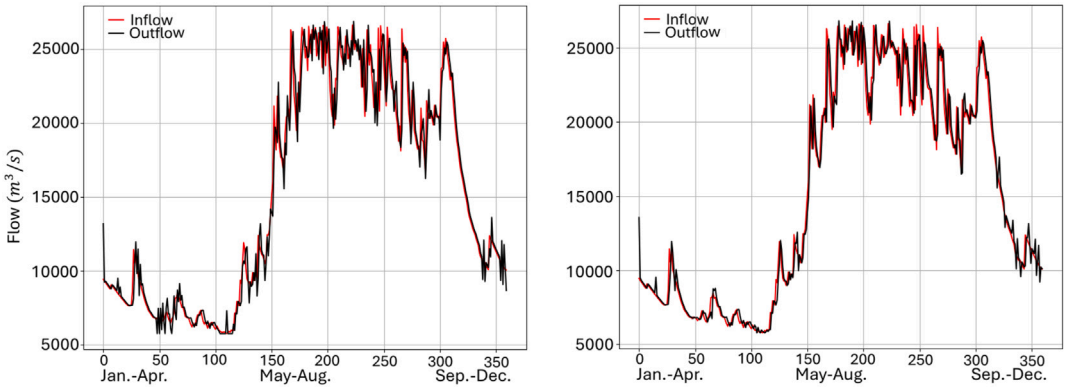


Fig. B.26. Inflow and total outflow (extreme rainfall).

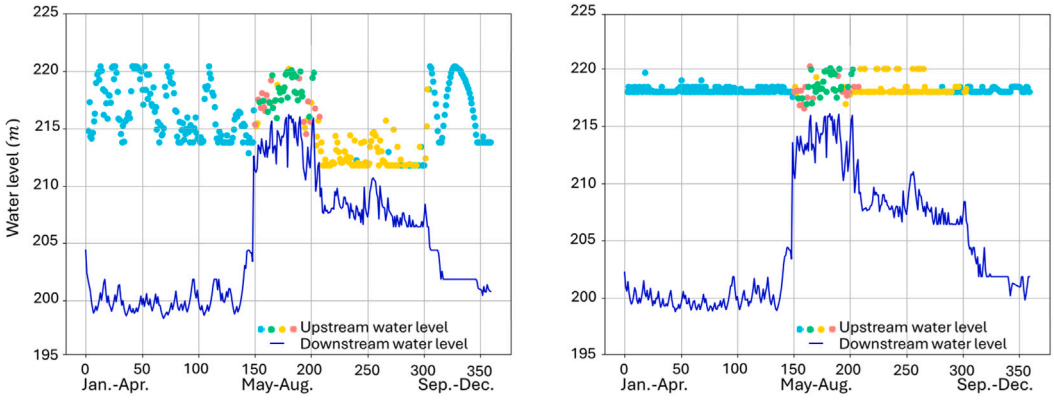


Fig. B.27. Upstream and downstream water levels (extreme inflow volatility).

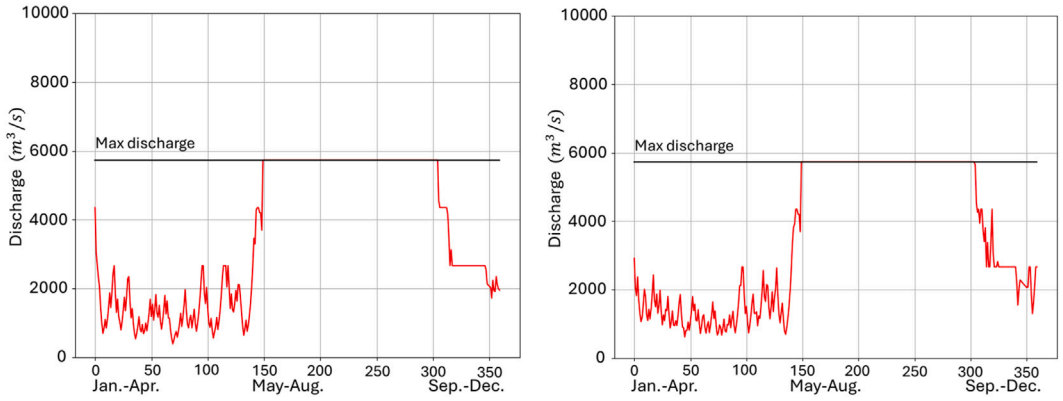


Fig. B.28. Discharge for hydropower (extreme inflow volatility).

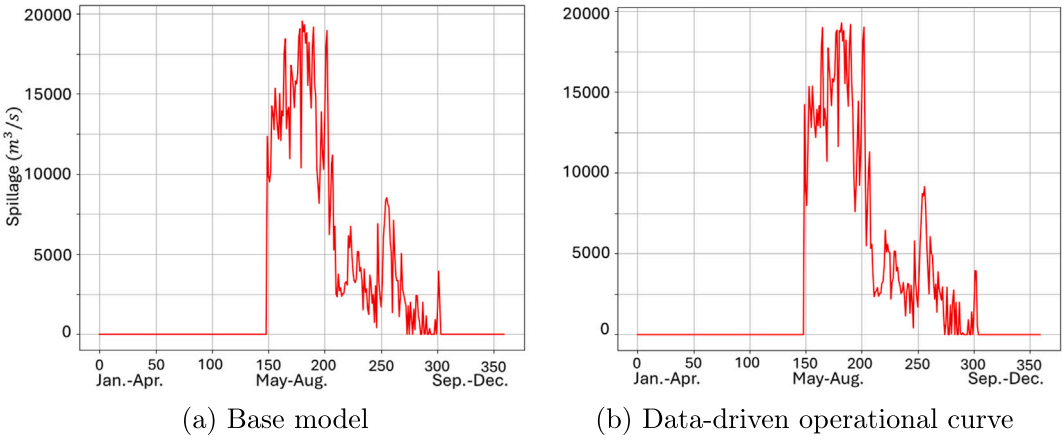


Fig. B.29. Spillage (extreme inflow volatility).

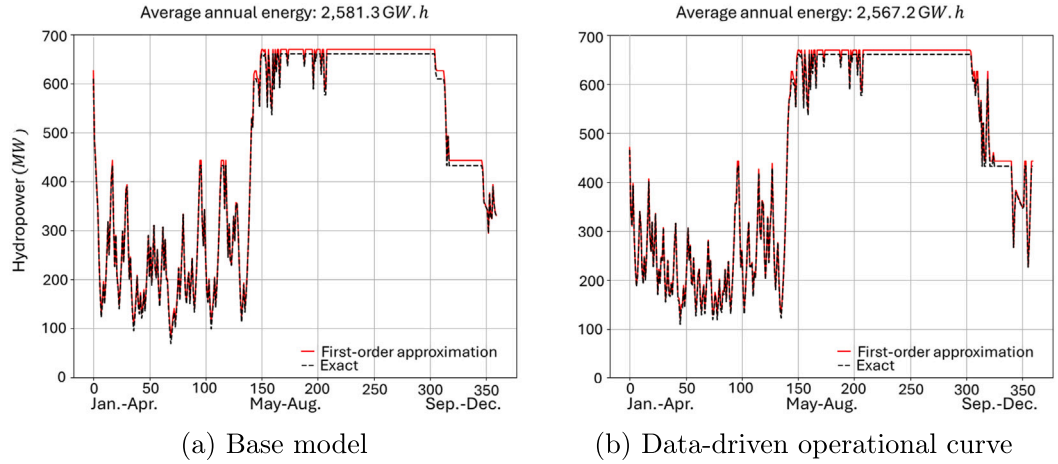


Fig. B.30. Hydropower produced by the reservoir (extreme inflow volatility).

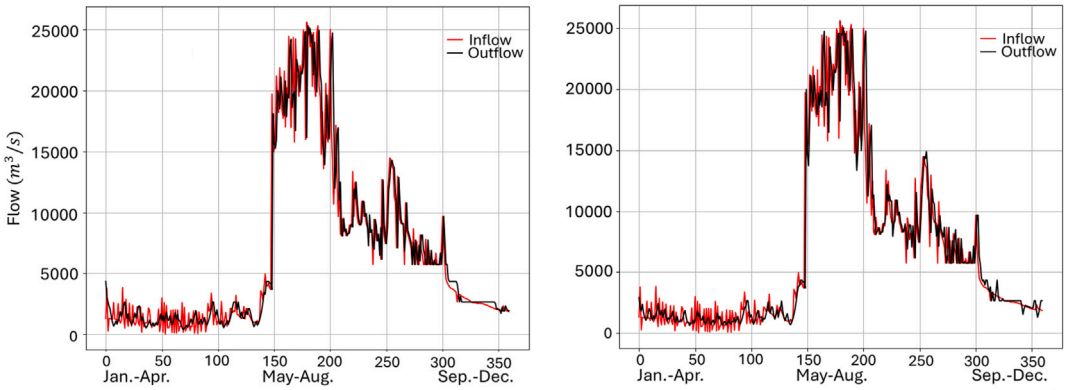


Fig. B.31. Inflow and total outflow (extreme inflow volatility).

curves across a broad range of realistic system conditions, particularly when there is evidence of shifts in key climate parameters.

Data availability

Data will be made available on request.

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